

Design and Development of an AI-Driven Personalized Career Guidance System Using NLP and Recommender Systems

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By

Shilbhushan Siddhartha Khanderao

Roll No.: 20245193

Supervisor:

Dr. Amit Kumar Upadhyay



Department of Computer Science and Engineering

Institute of Engineering & Technology

MANGALAYATAN UNIVERSITY

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APPROVAL SHEET

This dissertation entitled "**Design and Development of an AI-Driven Personalized Career Guidance System Using NLP and Recommender Systems**" by **Shilbhushan S. Khanderao (Roll No. 20245193)** is approved for the degree of Master of Technology in Computer Science (AI & ML).

Examiners

Supervisor(s)

Chairman

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(Name & Signature)

Dr. Amit Kumar Upadhyay

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(Signature)

Shilbhushan S. Khanderao

Roll no.: 20245193

Date: 11/08/26

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ABSTRACT

Career decision-making is a critical process for students, yet traditional counselling methods often rely on subjective human interpretation, limited expert availability, and rigid questionnaire-based evaluations. These conventional approaches struggle to scale to large student populations, frequently fail to capture the complexity and diversity of individual aspirations expressed in free-form language, and offer little transparency into how a given recommendation was reached. With the growing complexity of modern career pathways, the proliferation of unstructured student data, and the rapid diversification of occupational domains, there is a pressing need for scalable, intelligent systems capable of understanding individual interests expressed in natural language and translating them into personalized, explainable career recommendations.

This dissertation presents the design, development, and evaluation of an AI-driven career guidance system that integrates Natural Language Processing (NLP), Machine Learning (ML), transformer-based semantic modelling, and hybrid recommendation techniques to analyze free-form user input and suggest suitable career pathways. The system employs TF-IDF vectorization combined with a Logistic Regression classifier for career-domain prediction, Sentence-BERT (SBERT) embeddings for contextual semantic similarity computation, and K-Means clustering for uncovering natural groupings within user interest patterns. A hybrid scoring mechanism combines semantic similarity, keyword-based matching, Big Five personality-trait alignment, and rule-based domain logic to produce explainable Top-5 career recommendations, allowing the system to justify each suggestion rather than functioning as an opaque black box.

The system was implemented as a modular pipeline spanning text preprocessing, career-domain classification, semantic similarity analysis, personality-aware scoring, and clustering-based exploratory analysis, all integrated behind a Streamlit-based web interface that enables real-time interaction and displays domain predictions, similarity scores, and detailed recommendation breakdowns to the end user. The system was trained and evaluated using a career-domain dataset spanning 48 occupational categories and a curated 767-entry career-profile knowledge base, and the resulting hybrid framework was validated through comparative analysis against baseline TF-IDF and standalone SBERT approaches.

Experimental results demonstrate strong domain-classification accuracy (99.30%), measurably superior semantic understanding through SBERT embeddings compared to purely keyword-based methods (only 32% recommendation overlap between the two approaches), coherent and interpretable clustering behavior validated through silhouette analysis, and consistently high relevance in the generated Top-5 recommendations. The hybrid scoring approach was shown to combine these signals into recommendations that are both more broadly relevant and, unlike a single-technique baseline, fully explainable to the end user. The findings validate the effectiveness of integrating classical machine learning with transformer-based semantic models for educational decision support, and demonstrate that explainability need not come at the cost of recommendation quality.

The proposed system offers a scalable, modular framework suitable for institutional deployment and forms a foundation for future enhancements such as deeper psychometric integration, multilingual support, real-time labor-market data incorporation, and deep-learning-based classification models. This work contributes to the growing field of AI-enabled career counselling, demonstrating its potential to complement — rather than replace — traditional guidance practices with data-driven, personalized, and explainable insights.

Layout of the Thesis: Chapter 1 (Introduction) justifies the problem, defines the scope of the investigation, and outlines the motivation, objectives, and overall organization of the dissertation. Chapter 2 (Literature Review) critically reviews existing work on career guidance systems, Natural Language Processing, Machine Learning, transformer models, recommender systems, and clustering techniques, and identifies the research gap addressed by this study. Chapter 3 (System Design and Methodology) describes the proposed system architecture, research methodology, mathematical foundations, algorithms, and overall workflow of the AI-driven career guidance system. Chapter 4 (Implementation) documents the software implementation of the proposed design. Chapter 5 (Results and Discussion) presents the experimental evaluation, comparative analysis, and critical discussion of the findings against the research questions and literature. Chapter 6 (Summary, Conclusions, and Future Scope) summarizes the research findings, highlights the major contributions, discusses limitations, and outlines directions for future work.

Key Words: Career Guidance System, Natural Language Processing, Machine Learning, Sentence-BERT, TF-IDF, Logistic Regression, K-Means Clustering, Hybrid Recommendation, Explainable AI, Personality-Aware Recommendation.

TABLE OF CONTENTS

CHAPTER 1: INTRODUCTION.....	1
1.1 Background.....	1
1.2 Problem Statement.....	5
1.3 Research Questions	7
1.4 Research Objectives	8
1.5 Scope of the Work.....	9
1.5.1 Functional Scope	9
1.5.2 Research Scope.....	10
1.5.3 Out of Scope	10
1.6 Significance of the Study.....	11
1.6.1 Academic Significance	11
1.6.2 Technological Significance	11
1.6.3 Educational Significance	12
1.6.4 Practical Significance	12
1.7 Contributions of the Study.....	12
1.8 Organization of the Dissertation.....	14
CHAPTER 2: LITERATURE REVIEW.....	15
2.1 Introduction	15
2.2 Evolution of Career Guidance Systems.....	17
2.3 Traditional Career Counselling Approaches	21
2.4 Natural Language Processing for Career Guidance	24
2.5 Machine Learning Techniques for Text Classification	28
2.6 Transformer Models and Semantic Representation	31
2.7 Personality Modelling in AI-based Career Recommendation.....	34
2.8 Recommender Systems and Hybrid Recommendation Approaches	37

2.9 Clustering and Unsupervised Learning for Interest Grouping	40
2.10 Existing AI-based Career Guidance Systems	43
2.11 Research Gap Analysis	47
2.12 Summary	50
CHAPTER 3: SYSTEM DESIGN AND METHODOLOGY	53
3.1 Introduction	53
3.2 Overall System Architecture	53
3.3 Requirement Analysis	55
3.3.1 Functional Requirements	56
3.3.2 Non-Functional Requirements	56
3.3.3 System Constraints	57
3.4 Dataset Description	57
3.4.1 Domain Classification Dataset	58
3.4.2 Career Knowledge Dataset	58
3.4.3 Dataset Preparation Process	59
3.5 Semantic Similarity Module (SBERT)	60
3.5.1 Motivation for Using SBERT	60
3.5.2 Semantic Similarity Computation	60
3.5.3 Cosine Similarity Measurement	60
3.5.4 Integration within the Recommendation Framework	61
3.6 K-Means Clustering (Unsupervised Learning)	61
3.6.1 Motivation for Using K-Means Clustering	62
3.6.2 Clustering Methodology	62
3.6.3 Cluster Visualization Using Principal Component Analysis	63
3.6.4 Role of Clustering within the Proposed Framework	63
3.7 Personality Trait Inference	64
3.7.1 Big Five Personality Framework	64

3.7.2 Personality Inference Process	65
3.7.3 Contribution to Recommendation Generation.....	66
3.8 Hybrid Recommendation Engine	66
3.8.1 Components of the Hybrid Recommendation Engine.....	67
3.8.2 Hybrid Scoring Model.....	67
3.8.3 Recommendation Generation Process	68
3.9 System Workflow Summary	69
3.10 Summary.....	70
CHAPTER 4: IMPLEMENTATION	71
4.1 Introduction	71
4.2 Development Environment.....	71
4.3 Dataset Preparation.....	72
4.3.1 Domain Classification Dataset	73
4.3.2 Career Knowledge Dataset	73
4.3.3 Preprocessing Pipeline.....	73
4.4 Domain Classification Module.....	74
4.5 Semantic Similarity Module.....	75
4.6 Hybrid Recommendation Engine	76
4.7 Clustering Module	77
4.8 User Interface	78
4.9 Summary.....	78
CHAPTER 5: RESULTS AND DISCUSSION	79
5.1 Introduction	79
5.2 Domain Classification Performance.....	79
5.3 Semantic Similarity: TF-IDF versus SBERT	81
5.4 Hybrid Recommendation Output.....	84
5.5 Clustering Results.....	85

5.6 Recommendation Relevance	89
5.7 Consolidated Summary of Experimental Findings.....	89
5.8 Discussion.....	90
5.9 Threats to Validity	91
5.10 Summary.....	92
CHAPTER 6: SUMMARY, CONCLUSIONS, AND FUTURE SCOPE	93
6.1 Introduction	93
6.2 Summary of Work	93
6.3 Major Research Contributions.....	94
6.4 Conclusions	96
6.5 Limitations of the Present Study	97
6.6 Future Scope	99
6.7 Concluding Remarks	101
LITERATURE CITED.....	103
APPENDIX I -- Sample User Inputs for System Evaluation	108
APPENDIX II -- Key Implementation Code Snippets.....	110
II.1 NLP Text Preprocessing	110
II.2 TF-IDF and Logistic Regression Domain Classifier	110
II.3 SBERT Semantic Similarity Module.....	110
II.4 K-Means Clustering	111
APPENDIX III -- Dataset Structure	112
III.1 Domain Classification Dataset	112
III.2 Career Recommendation Dataset	112
APPENDIX IV -- User Interface Screenshots.....	114
IV.1 Application Home Screen	114
IV.2 Domain Prediction Output	114
IV.3 Top-5 Career Recommendations.....	115

APPENDIX V -- Software Requirements and Application Execution	116
V.1 Software Requirements.....	116
V.2 Application Execution	116
APPENDIX VI -- Validation and Test Cases.....	117

LIST OF FIGURES

Figure No.	Title	Page No.
Figure 1.1	Motivation for AI-Driven Career Guidance Amid Evolving Educational and Workforce Demands	2
Figure 2.1	Stages in the Evolution of Career Guidance Systems: From Traditional Counselling to AI-Driven Hybrid Recommendation	20
Figure 2.2	Research Gap Leading to the Proposed Framework	49
Figure 3.1	Overall System Architecture of the Proposed AI-Driven Career Guidance System	54
Figure 3.2	Dataset Organization	58
Figure 3.3	Dataset Preparation Workflow	59
Figure 3.4	Semantic Similarity Workflow	61
Figure 3.5	K-Means Clustering Workflow	63
Figure 3.6	Personality Inference Process	65
Figure 3.7	Architecture of the Hybrid Recommendation Engine	67
Figure 3.8	Recommendation Generation Pipeline	68
Figure 3.9	End-to-End Workflow of the Proposed AI-Driven Career Guidance System	69
Figure 5.1	Classification Performance Summary of the Domain Classification Model	80
Figure 5.2	Confusion Matrix of the Domain Classification Model	80
Figure 5.3	Experimental Comparison of TF-IDF and SBERT Similarity Scores	82
Figure 5.4	Overlap Between TF-IDF and SBERT Top-Ranked Recommendations	82
Figure 5.5	Silhouette Score versus Number of Clusters ($k = 2$ to 15)	86

Figure 5.6	TF-IDF K-Means Clusters ($k = 8$), Colored by Cluster and by True Domain	87
Figure III.1	Domain Dataset Distribution Diagram	113
Figure IV.1	Application Home Screen	114
Figure IV.2	Domain Prediction Output	114
Figure IV.3	Top-5 Career Recommendations	115

LIST OF TABLES

Table No.	Title	Page No.
Table 1.1	Major Challenges in Existing Career Guidance Systems	5
Table 1.2	Major Contributions of the Proposed Study	12
Table 1.3	Organization of the Dissertation	14
Table 2.1	Comparative Overview of Career Guidance System Generations by Approach, Strengths, and Limitations	20
Table 2.2	Comparison of Traditional Career Counselling Frameworks	24
Table 2.3	Natural Language Processing Pipeline for Career Recommendation	27
Table 2.4	Overview of Supervised Machine Learning Algorithms for Text Classification	30
Table 2.5	Evolution of Language Representation Techniques	34
Table 2.6	Comparative Analysis of Recommendation Approaches	39
Table 2.7	Comparison of Clustering Algorithms	42
Table 2.8	Comparative Analysis of Existing AI-based Career Guidance Systems	46
Table 2.9	Research Gap Analysis	49
Table 3.1	Architecture Layers of the Proposed System	54
Table 3.2	Functional Requirements of the Proposed System	56
Table 3.3	Non-Functional Requirements	56
Table 3.4	Characteristics of the Domain Classification Dataset	58
Table 3.5	Structure of the Career Knowledge Dataset	59
Table 3.6	Configuration of the K-Means Clustering Model	64
Table 3.7	Big Five Personality Traits Used in the Proposed System	65
Table 3.8	Weight Distribution of the Hybrid Recommendation Model	68
Table 3.9	Summary of System Workflow	69

Table 4.1	Software Tools and Libraries Used for System Implementation	71
Table 4.2	Datasets Used During Experimental Evaluation	72
Table 4.3	Preprocessing Operations Performed Before Model Training	74
Table 4.4	Experimental Configuration of the Domain Classification Model	74
Table 4.5	Configuration of the Sentence-BERT Embedding Module	75
Table 4.6	Recommendation Components Used in the Hybrid Recommendation Framework	76
Table 4.7	Implementation Parameters of the K-Means Clustering Module	77
Table 5.1	Performance Evaluation Metrics of the Domain Classification Model	79
Table 5.2	Quantitative Comparison Between TF-IDF and SBERT Similarity	82
Table 5.3	Semantic Comparison between TF-IDF and Sentence-BERT	83
Table 5.4	Computational Comparison between TF-IDF and Sentence-BERT	84
Table 5.5	Sample Hybrid Recommendation Score Breakdown	84
Table 5.6	Summary of Generated Clusters (k = 8, TF-IDF)	86
Table 5.7	Summary of Experimental Clustering Results	87
Table 5.8	Comparison of Recommendation Relevance between TF-IDF-Only and the Full Hybrid (SBERT-Based) Engine	89
Table 5.9	Summary of Experimental Findings	90
Table 6.1	Summary of Major Research Activities	94
Table 6.2	Summary of Major Research Contributions	95
Table 6.3	Achievement of Research Objectives	97
Table 6.4	Summary of Research Limitations and Their Implications	98
Table 6.5	Potential Future Enhancements to the Proposed Career Guidance Framework	101

Table I.1	Representative User Inputs Used for System Evaluation	108
Table III.1	Domain Classification Dataset Summary	112
Table III.2	Sample Training Records	112
Table VI.1	Module-Level Validation Checks	117
Table VI.2	System-Level Test Scenarios	118

ABBREVIATIONS AND NOMENCLATURE

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
BERT	Bidirectional Encoder Representations from Transformers
CF	Collaborative Filtering
DBSCAN	Density-Based Spatial Clustering of Applications with Noise
EDM	Educational Data Mining
FFM	Five Factor Model
GMM	Gaussian Mixture Model
IDF	Inverse Document Frequency
IEEE	Institute of Electrical and Electronics Engineers
L-BFGS	Limited-memory Broyden–Fletcher–Goldfarb–Shanno (optimization algorithm)
LSTM	Long Short-Term Memory
ML	Machine Learning
NLP	Natural Language Processing
NLTK	Natural Language Toolkit
PCA	Principal Component Analysis
RIASEC	Realistic, Investigative, Artistic, Social, Enterprising, Conventional (Holland's Model)
RNN	Recurrent Neural Network
ROC	Receiver Operating Characteristic
RS	Recommender System
SBERT	Sentence-BERT
STS	Semantic Textual Similarity
SVM	Support Vector Machine

Abbreviation	Full Form
TF-IDF	Term Frequency–Inverse Document Frequency
t-SNE	t-distributed Stochastic Neighbor Embedding
UI	User Interface
XAI	Explainable Artificial Intelligence
XR	Extended Reality

CHAPTER 1: INTRODUCTION

1.1 Background

Career selection is one of the most significant decisions in an individual's academic and professional journey, as it directly influences future educational opportunities, professional growth, financial stability, and personal satisfaction. Selecting a career that aligns with an individual's interests, abilities, personality, and long-term aspirations is essential for achieving professional success and job satisfaction. However, the rapid evolution of technology, emergence of interdisciplinary career domains, and continuously changing industry requirements have made career decision-making increasingly complex. Students today are presented with thousands of career options spanning conventional professions as well as emerging fields such as Artificial Intelligence, Data Science, Cybersecurity, Cloud Computing, Extended Reality (XR), Robotics, Bioinformatics, and Digital Marketing. Consequently, identifying the most suitable career pathway has become a challenging task, particularly for students with limited exposure to industry trends and professional guidance.

Traditionally, career counselling has relied on human counsellors, aptitude assessments, psychometric tests, structured interviews, and standardized questionnaires to evaluate students' interests and recommend suitable career options. Although these approaches have been widely adopted by educational institutions, they present several practical limitations. The quality of recommendations often depends on the expertise and experience of the counsellor, making the process subjective and inconsistent. Furthermore, conducting individual counselling sessions for large student populations is both time-consuming and resource-intensive. Many institutions also face a shortage of trained career counsellors, resulting in high student-to-counsellor ratios that restrict the availability of personalized guidance. As a result, many students rely on informal sources such as social media, peer opinions, or incomplete online information, which may not accurately reflect their interests or capabilities.

The rapid advancement of Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP) has significantly transformed educational decision-support systems by enabling scalable, data-driven, and personalized recommendation

frameworks. AI-powered systems can analyze large volumes of structured and unstructured educational data, identify complex relationships between user preferences and career profiles, and generate recommendations that are more consistent than traditional expert-driven approaches. Recent developments in Educational Artificial Intelligence (AIED) have demonstrated the potential of intelligent recommendation systems to support academic advising, learning analytics, and career planning through adaptive and context-aware decision-making mechanisms [43, 9].

The growing adoption of Industry 4.0 technologies, digital transformation, and automation has fundamentally changed workforce requirements across industries. According to recent reports published by international organizations such as the World Economic Forum, the demand for digital competencies, analytical thinking, Artificial Intelligence, cloud computing, cybersecurity, and interdisciplinary skills continues to increase, while several traditional job roles are undergoing rapid transformation. Consequently, career guidance systems must continuously adapt to changing labor market requirements and provide recommendations that align with both individual interests and evolving industry demands [54].

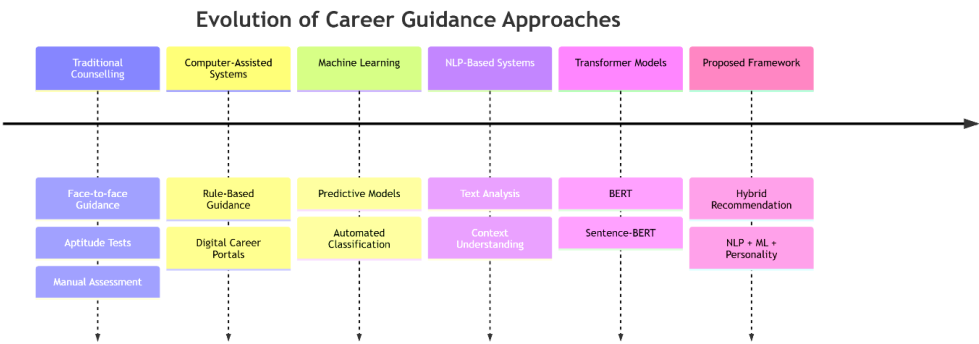


Figure 1.1 Motivation for AI-Driven Career Guidance Amid Evolving Educational and Workforce Demands

Natural Language Processing (NLP) has become one of the fundamental technologies supporting intelligent information retrieval, text analytics, recommendation systems, and conversational Artificial Intelligence. By enabling computers to interpret and analyze human language, NLP facilitates the extraction of meaningful information from unstructured textual data, making it particularly suitable for educational decision-support applications [1, 6]. Modern NLP techniques enable computers to process free-form textual

input, extract meaningful information, and identify underlying user intent without requiring rigid questionnaire-based interactions. Classical text processing techniques such as tokenization, stop-word removal, stemming, lemmatization, and Term Frequency–Inverse Document Frequency (TF-IDF) vectorization provide effective mechanisms for transforming textual information into machine-readable representations suitable for machine learning algorithms. These techniques have demonstrated strong performance in various text classification tasks, including document categorization, sentiment analysis, spam detection, and information retrieval.

Machine Learning algorithms further enhance the capability of NLP systems by learning discriminative patterns from labelled datasets. Among the widely adopted supervised learning algorithms, Logistic Regression has consistently demonstrated excellent performance on high-dimensional sparse textual data generated using TF-IDF representations. Owing to its computational efficiency, probabilistic interpretation, robustness on high-dimensional sparse feature spaces, and ease of implementation, Logistic Regression continues to serve as a strong baseline classifier for numerous Natural Language Processing applications, including document classification, spam detection, sentiment analysis, and educational text mining [13, 15]. Within the context of career guidance, domain classification enables the system to automatically identify the broad career category corresponding to a user's expressed interests, thereby reducing the search space for subsequent recommendation stages.

While traditional feature-based approaches effectively identify keyword occurrences, they often fail to capture contextual meaning and semantic relationships between words and sentences. Two users may express identical career interests using entirely different vocabulary, making keyword-based systems insufficient for generating accurate recommendations. The introduction of transformer-based language models has significantly improved semantic understanding by learning contextual representations of natural language. Bidirectional Encoder Representations from Transformers (BERT) and its optimized variant, Sentence-BERT (SBERT), generate dense semantic embeddings that preserve contextual relationships between sentences rather than relying solely on lexical similarity. These semantic representations enable AI systems to recognize similar intentions even when different words or sentence structures are used, thereby improving recommendation quality and user experience. Compared with traditional vector-space

models, transformer-based semantic representations have consistently demonstrated superior performance in semantic textual similarity, information retrieval, and recommendation tasks by preserving contextual relationships between sentences rather than relying solely on lexical overlap [4, 3].

In addition to semantic understanding, behavioral and psychological characteristics play an important role in determining career suitability. Personality assessment models, particularly the widely accepted Big Five Personality Traits framework, provide valuable insights into behavioral tendencies that influence career preferences, workplace performance, and long-term job satisfaction. Integrating lightweight personality inference with semantic similarity analysis enables recommendation systems to move beyond simple keyword matching and generate recommendations that better align with individual behavioral characteristics. Such integration improves personalization while maintaining system explainability. Several recent studies have reported that incorporating behavioral characteristics into recommendation systems improves recommendation relevance, user engagement, and long-term satisfaction, particularly in educational and career guidance applications [17, 35].

Unsupervised learning techniques also contribute significantly to educational recommendation systems by discovering hidden structures within data. Clustering algorithms such as K-Means can automatically group users with similar interest patterns, enabling better visualization, exploratory analysis, and interpretation of career domains. When combined with dimensionality reduction techniques such as Principal Component Analysis (PCA), clustering provides meaningful insights into the underlying structure of user interests and facilitates improved understanding of recommendation behavior. The combination of clustering and dimensionality reduction techniques has also become a common practice for exploratory data analysis and visualization in educational analytics and recommendation systems [7, 20].

Explainability has emerged as a critical requirement for Artificial Intelligence systems deployed in high-impact domains such as healthcare, finance, and education, where automated decisions directly influence human outcomes. Since career recommendations can significantly affect an individual's academic and professional future, users must understand the reasoning behind the generated recommendations to establish trust and confidence in the system [14, 55]. Since career recommendations can substantially

influence an individual's educational and professional decisions, users must understand why a particular recommendation has been generated. Transparent recommendation mechanisms increase user confidence, improve trust, and support informed decision-making. Hybrid recommendation approaches that combine semantic similarity, keyword matching, personality alignment, and rule-based reasoning provide more interpretable recommendations compared to purely black-box models while maintaining high predictive performance.

Motivated by these developments, this dissertation presents the design and development of an AI-driven personalized career guidance system that integrates Natural Language Processing, Machine Learning, transformer-based semantic modelling, personality-aware recommendation, and unsupervised learning into a unified recommendation framework. The proposed system accepts free-form textual descriptions of user interests, predicts the most relevant career domain using TF-IDF and Logistic Regression, measures semantic similarity through Sentence-BERT embeddings, identifies meaningful interest patterns using K-Means clustering, and generates explainable Top-5 career recommendations using a hybrid scoring mechanism. By integrating multiple complementary AI techniques within a modular architecture, the proposed system aims to provide scalable, accurate, transparent, and personalized career guidance suitable for educational institutions and individual learners.

1.2 Problem Statement

Career guidance has become increasingly challenging due to the rapid evolution of technology, diversification of professional roles, and changing industry skill requirements. Students are often required to choose career paths from a vast number of available options without possessing sufficient knowledge about industry expectations, emerging technologies, or their own strengths and interests. As a result, many students experience uncertainty while selecting suitable academic specializations and career pathways, leading to inappropriate career choices, skill mismatches, and reduced professional satisfaction.

Table 1.1 Major Challenges in Existing Career Guidance Systems

Challenge	Impact
Manual counselling	Limited scalability

Challenge	Impact
Fixed questionnaires	Poor personalization
Keyword matching	Weak semantic understanding
No personality modelling	Generic recommendations
Limited explainability	Reduced user trust
Separate AI models	Low recommendation quality

Conventional career counselling practices primarily depend on structured questionnaires, aptitude tests, psychometric assessments, and expert counsellor interactions. Although these approaches have been successfully adopted for many years, they exhibit several inherent limitations. First, they require considerable human involvement, making them difficult to scale for institutions with large student populations. Second, the recommendations often depend upon subjective interpretation by counsellors, resulting in variations in decision quality and consistency. Third, most traditional counselling systems require users to answer predefined questions, limiting their ability to capture nuanced interests, aspirations, and motivations that are naturally expressed through free-form language.

The emergence of online career guidance platforms has partially addressed accessibility issues; however, many existing digital systems continue to rely on keyword-based search mechanisms or rule-based recommendation engines. Such systems primarily focus on lexical similarity and often fail to understand the contextual meaning of user statements. Consequently, users expressing identical interests using different vocabulary may receive inconsistent or irrelevant recommendations. Furthermore, many existing systems generate recommendations without providing sufficient explanation regarding the reasoning behind the suggested career paths, thereby reducing user confidence and trust in the recommendation process.

Another significant limitation observed in existing career recommendation systems is the isolated use of Artificial Intelligence techniques. Numerous approaches employ traditional machine learning algorithms for domain classification, while others utilize semantic similarity models or rule-based filtering independently. Very few systems integrate multiple complementary techniques such as supervised learning, semantic embeddings,

personality-aware analysis, and unsupervised clustering within a unified recommendation framework. The absence of such integration restricts the ability of existing systems to provide highly personalized, context-aware, and explainable career recommendations.

Furthermore, personality characteristics significantly influence career preferences, workplace behavior, and long-term job satisfaction. Despite their importance, personality-related information is often ignored or treated as an optional component in existing recommendation systems. Similarly, most available solutions lack mechanisms for discovering hidden patterns among users with similar interests, thereby limiting opportunities for exploratory analysis and improved recommendation transparency.

These limitations highlight the need for an intelligent, scalable, and explainable career guidance framework capable of understanding natural language descriptions, analyzing semantic intent, incorporating behavioral characteristics, and producing personalized recommendations supported by transparent decision-making mechanisms.

Therefore, this dissertation proposes the design and development of an AI-driven personalized career guidance system that integrates Natural Language Processing, Machine Learning, transformer-based semantic modelling, personality-aware recommendation, and clustering techniques into a unified hybrid recommendation framework. The proposed system aims to overcome the limitations of existing approaches by accurately classifying user interests, understanding contextual semantics, identifying meaningful interest patterns, and generating explainable Top-5 career recommendations tailored to individual users.

1.3 Research Questions

Based on the identified research problem and the limitations of existing career guidance systems, this study aims to investigate the following research questions:

RQ1. Can Natural Language Processing techniques accurately interpret free-form user descriptions and classify them into appropriate career domains?

RQ2. Does the integration of transformer-based semantic embeddings (SBERT) improve recommendation quality compared to traditional keyword-based similarity methods?

RQ3. Can the incorporation of personality-related information enhance the relevance and personalization of career recommendations?

RQ4. How effectively can a hybrid recommendation framework combining domain classification, semantic similarity, personality inference, and rule-based scoring generate explainable career recommendations?

RQ5. Can clustering techniques identify meaningful patterns in user interests that improve system interpretability and recommendation analysis?

The answers to these research questions guide the overall design, implementation, and evaluation of the proposed AI-driven career guidance system presented in this dissertation.

1.4 Research Objectives

The primary objective of this research is to design, develop, and evaluate an Artificial Intelligence-driven personalized career guidance system capable of understanding users' interests expressed in natural language and generating accurate, explainable, and personalized career recommendations. The proposed system aims to integrate multiple Artificial Intelligence techniques within a unified framework to overcome the limitations of conventional career counselling approaches and existing recommendation systems.

The specific objectives of this research are as follows:

To investigate the effectiveness of Natural Language Processing techniques for analyzing free-form textual descriptions provided by users and transforming them into structured representations suitable for machine learning applications.

To develop a supervised machine learning model based on TF-IDF feature extraction and Logistic Regression for accurately classifying user interests into predefined career domains.

To evaluate the effectiveness of transformer-based semantic embedding models, particularly Sentence-BERT (SBERT), in measuring semantic similarity between user interests and career profiles.

To design and implement a hybrid recommendation framework that combines semantic similarity, keyword matching, personality-aware analysis, and rule-based scoring to generate accurate and explainable career recommendations.

To investigate the application of unsupervised learning techniques, specifically K-Means clustering, for identifying meaningful patterns in user interests and improving the interpretability of recommendation outcomes.

To develop an interactive web-based application using the Streamlit framework that enables users to obtain personalized career recommendations through an intuitive and user-friendly interface.

To evaluate the performance of the proposed system using standard machine learning evaluation metrics, semantic similarity analysis, clustering visualization, and recommendation quality assessment.

To demonstrate the feasibility of integrating multiple Artificial Intelligence techniques into a scalable educational decision-support system capable of assisting students in career planning and decision-making.

The successful achievement of these objectives establishes the technical and research foundation for developing an intelligent career guidance system capable of providing personalized, context-aware, and explainable career recommendations.

1.5 Scope of the Work

The scope of this dissertation encompasses the design, development, implementation, and evaluation of an Artificial Intelligence-driven career guidance system that integrates Natural Language Processing, Machine Learning, semantic modelling, and recommendation techniques within a unified framework. The proposed system focuses on assisting students and learners in identifying suitable career pathways based on their interests, skills, and behavioral characteristics expressed through natural language.

1.5.1 Functional Scope

The functional scope of the proposed system includes the following components:

Development of an NLP preprocessing pipeline for text normalization, tokenization, stop-word removal, and feature preparation.

Implementation of TF-IDF-based feature extraction and Logistic Regression for multi-class career domain classification.

Integration of Sentence-BERT (SBERT) embeddings to perform semantic similarity analysis between user inputs and career profiles.

Application of K-Means clustering to identify natural groupings of user interests and support exploratory analysis.

Development of a hybrid recommendation engine combining semantic similarity, keyword overlap, personality inference, and rule-based scoring.

Design of a web-based user interface using Streamlit to enable real-time interaction and recommendation generation.

1.5.2 Research Scope

From a research perspective, this work investigates the effectiveness of combining classical machine learning techniques with modern transformer-based language models for personalized career recommendation. The study also evaluates the contribution of semantic understanding, clustering, and personality-aware recommendation in improving recommendation relevance and explainability.

The implementation is restricted to English-language textual input and utilizes curated datasets consisting of career profiles and domain-labelled training samples. The recommendation engine is intended to support educational career guidance and is not designed to replace professional counselling services.

1.5.3 Out of Scope

The following aspects are beyond the scope of the present investigation:

- Integration with real-time employment portals and labor market analytics.
- Comprehensive psychometric assessments using standardized psychological instruments.
- Multilingual recommendation support.
- Fine-tuning of large language models for career prediction.
- Mobile application development and cloud-native deployment.
- Continuous learning from live user feedback.

These limitations provide opportunities for future research and system enhancement, which are discussed in Chapter 5.

1.6 Significance of the Study

1.6.1 Academic Significance

The increasing complexity of modern career landscapes has created a growing demand for intelligent systems capable of providing personalized, scalable, and data-driven career guidance. This research contributes to the growing field of Artificial Intelligence in education by demonstrating how Natural Language Processing, Machine Learning, semantic modelling, and recommendation systems can be effectively integrated to support career decision-making.

From an academic perspective, this dissertation presents a comprehensive framework that combines multiple Artificial Intelligence techniques within a unified recommendation architecture. Rather than relying on a single recommendation strategy, the proposed system integrates supervised learning, semantic similarity modelling, personality-aware analysis, clustering, and explainable hybrid scoring to generate more relevant and transparent career recommendations. This integration contributes to ongoing research in educational recommendation systems and intelligent decision-support applications.

1.6.2 Technological Significance

From a technological perspective, the proposed system demonstrates the practical applicability of combining classical machine learning approaches with transformer-based language models. The integration of TF-IDF-based domain classification with SBERT semantic embeddings illustrates how traditional statistical methods and modern deep learning representations can complement each other to improve recommendation accuracy while maintaining computational efficiency and interpretability.

The study also contributes to the development of explainable recommendation systems. Unlike conventional recommendation engines that often function as black-box models, the proposed hybrid scoring mechanism provides transparent reasoning by combining semantic similarity, keyword matching, personality alignment, and rule-based factors. Such transparency increases user trust and enables students to better understand the rationale behind the recommended career options.

1.6.3 Educational Significance

From an educational perspective, the proposed system has the potential to support schools, colleges, universities, and career counselling centres by providing scalable and consistent guidance to large numbers of students. The ability to process free-form textual input reduces dependence on rigid questionnaires while enabling more natural interaction between users and the recommendation system.

1.6.4 Practical Significance

The proposed framework also has practical significance for educational technology platforms, online learning environments, and career counselling applications that seek to incorporate Artificial Intelligence into their recommendation processes. Its modular architecture enables future integration with psychometric assessments, labor market information systems, multilingual NLP models, and cloud-based educational services.

Overall, this research demonstrates how Artificial Intelligence can be effectively employed to enhance career guidance by improving personalization, recommendation quality, transparency, and scalability, thereby supporting better educational and professional decision-making.

1.7 Contributions of the Study

The major contributions of this dissertation are summarized as follows:

Table 1.2: Major Contributions of the Proposed Study

Contribution	Description
Unified AI-driven Career Guidance Framework	Developed an integrated career guidance framework that combines Natural Language Processing (NLP), supervised Machine Learning (ML), semantic embeddings, clustering, and hybrid recommendation techniques to generate personalized and explainable career recommendations.

Contribution	Description
Multi-class Career Domain Classification	Designed and implemented a supervised classification model using TF-IDF feature extraction and Logistic Regression for accurate prediction of career domains from free-form textual user inputs.
Semantic Understanding using Sentence-BERT	Integrated Sentence-BERT (SBERT) embeddings to capture contextual semantic relationships between user interests and career profiles, enabling more meaningful recommendation generation beyond traditional keyword matching.
Hybrid Recommendation Mechanism	Proposed an explainable hybrid recommendation model that combines semantic similarity, keyword matching, personality-aware analysis, and rule-based scoring to generate personalized Top-5 career recommendations with improved transparency.
Interest Pattern Discovery through Clustering	Applied the K-Means clustering algorithm to identify natural groupings of user interests, supporting exploratory analysis and improving the interpretability of recommendation outcomes.
Interactive Web-based Application	Developed a modular web application using the Streamlit framework that demonstrates the practical deployment of the proposed AI-driven career guidance system through an intuitive user interface.
Comprehensive Performance Evaluation	Evaluated the proposed framework using standard machine learning classification metrics, semantic similarity analysis, clustering visualization, and recommendation quality assessment to validate its effectiveness.

These contributions collectively demonstrate the feasibility of integrating multiple Artificial Intelligence techniques to develop an effective, scalable, and explainable career guidance system suitable for educational environments.

1.8 Organization of the Dissertation

The dissertation is structured as follows:

Table 1.3: Organization of the Dissertation

Chapter	Description
Chapter 1	Introduces the research background, problem statement, research questions, objectives, scope, significance, contributions, and overall organization of the dissertation.
Chapter 2	Reviews existing literature related to career guidance systems, Natural Language Processing, Machine Learning, transformer models, recommender systems, clustering techniques, and identifies the research gap addressed in this study.
Chapter 3	Describes the proposed system architecture, research methodology, mathematical foundations, algorithms, implementation strategy, and workflow of the proposed AI-driven career guidance system.
Chapter 4	Presents the implementation details, experimental setup, performance evaluation, comparative analysis, and discussion of experimental results.
Chapter 5	Summarizes the research findings, highlights the major contributions, discusses limitations, and outlines future research directions.

CHAPTER 2: LITERATURE REVIEW

2.1 Introduction

The rapid advancement of Artificial Intelligence (AI), Machine Learning (ML), and Natural Language Processing (NLP) has significantly transformed the development of intelligent decision-support systems across various domains, including healthcare, finance, education, and career guidance [22, 1]. In educational environments, the increasing demand for personalized learning experiences and data-driven academic advising has accelerated the adoption of AI-based recommendation systems capable of analyzing learner behavior, interests, and preferences to provide individualized guidance [22]. Among these applications, AI-driven career guidance systems have emerged as an important research area because they address the limitations of conventional counselling approaches by providing scalable, consistent, and personalized career recommendations.

Traditional career counselling methods primarily rely on psychometric assessments, aptitude tests, structured questionnaires, and expert counsellor interactions to evaluate an individual's interests, abilities, and personality traits [25, 27]. Although these approaches have proven effective in many educational settings, they are often constrained by subjectivity, limited scalability, and significant dependence on the expertise of professional counsellors. Furthermore, conventional counselling techniques typically require users to respond to predefined questionnaires, which may not adequately capture the complexity and diversity of individual aspirations expressed through natural language [38].

Recent developments in Natural Language Processing have enabled computers to understand and process unstructured textual information with remarkable accuracy. Classical text representation techniques such as Term Frequency–Inverse Document Frequency (TF-IDF) have been widely employed for document representation and text classification because of their computational efficiency and interpretability [2, 1]. However, these statistical approaches primarily rely on lexical information and often fail to capture contextual semantics, making them less effective when different users describe similar interests using different vocabulary.

To overcome these limitations, transformer-based language models have revolutionized semantic text representation by learning contextual embeddings that preserve the semantic

relationships between words and sentences [4, 3]. Models such as Bidirectional Encoder Representations from Transformers (BERT) and Sentence-BERT (SBERT) have demonstrated superior performance in semantic similarity, information retrieval, question answering, and recommendation systems by generating dense vector representations that capture contextual meaning rather than relying solely on keyword frequency [4, 3, 25]. These advances have significantly improved the capability of intelligent systems to interpret free-form user inputs and produce context-aware recommendations.

In parallel, supervised Machine Learning algorithms have become fundamental components of intelligent text analysis systems. Algorithms including Logistic Regression, Support Vector Machines (SVM), Random Forests, Decision Trees, and Naïve Bayes have been extensively applied to text classification problems due to their ability to learn discriminative patterns from labelled datasets [8, 9]. Within career recommendation systems, these classifiers facilitate automatic prediction of career domains, thereby reducing the search space for subsequent recommendation modules.

Another important research direction involves incorporating behavioral and psychological characteristics into recommendation systems. Personality assessment frameworks, particularly the Five-Factor Model (Big Five Personality Traits), have been widely adopted to understand behavioral tendencies that influence educational choices, workplace performance, and long-term career satisfaction [17, 35]. Recent studies have explored the integration of personality inference with Natural Language Processing techniques to generate recommendations that better align with individual behavioral characteristics, resulting in improved personalization and user satisfaction [35].

Furthermore, recommender systems have evolved considerably over the past two decades, progressing from traditional content-based and collaborative filtering approaches to sophisticated hybrid recommendation frameworks capable of combining multiple information sources [10, 25]. Hybrid recommendation strategies have gained widespread attention because they effectively address the limitations of individual recommendation techniques while improving recommendation accuracy, diversity, robustness, and explainability. Similarly, unsupervised learning methods such as K-Means clustering have been extensively employed to discover hidden structures within educational datasets, enabling exploratory analysis and improved visualization of user interest patterns [7, 20].

Despite these advancements, existing AI-driven career guidance systems continue to exhibit several limitations. Many systems rely exclusively on keyword matching or questionnaire-based approaches, while others focus only on semantic similarity or supervised classification without integrating complementary Artificial Intelligence techniques. Additionally, relatively few studies incorporate personality-aware recommendations, clustering-based exploratory analysis, and explainable hybrid scoring mechanisms within a unified recommendation framework [38, 39].

Therefore, a comprehensive review of existing literature is essential to understand the strengths, limitations, and research gaps associated with current AI-driven career guidance systems. This chapter critically examines previous research related to traditional career counselling, Natural Language Processing, Machine Learning, transformer-based semantic representation, personality modelling, recommender systems, clustering techniques, and existing career recommendation frameworks. The insights obtained from this review establish the theoretical foundation for the proposed AI-driven personalized career guidance system presented in the subsequent chapters.

2.2 Evolution of Career Guidance Systems

Career guidance has undergone significant transformation over the past century, evolving from traditional counsellor-led advisory services to intelligent Artificial Intelligence (AI)-driven recommendation systems. This evolution has been influenced by advancements in psychology, computer science, data analytics, and Artificial Intelligence, enabling career guidance systems to become increasingly personalized, scalable, and data-driven [23, 38].

The earliest career guidance practices emerged during the early twentieth century, when vocational counselling primarily depended on counsellors' expertise, structured interviews, and aptitude assessments. One of the pioneering contributions to vocational guidance was made by Frank Parsons, who proposed that career selection should be based on three fundamental aspects: understanding an individual's abilities, understanding occupational requirements, and matching personal characteristics with suitable professions [16]. Parsons' work laid the foundation for systematic career counselling and continues to influence modern career guidance methodologies.

As career counselling matured, psychological assessment models became an integral component of the decision-making process. Holland's Theory of Vocational Personalities and Work Environments (RIASEC) introduced a structured framework that categorizes both individuals and occupations into six personality types: Realistic, Investigative, Artistic, Social, Enterprising, and Conventional [15]. According to Holland's theory, individuals experience greater career satisfaction when their personality characteristics align with their occupational environment. Similarly, the Five-Factor Model (Big Five Personality Traits) has been widely adopted to evaluate behavioral tendencies associated with career preferences, workplace adaptability, leadership potential, and professional success [17, 35]. These psychological models remain important components of career counselling because they provide scientifically validated methods for understanding human behavior and vocational interests.

Although traditional counselling methods have proven effective in many educational environments, they possess several limitations. The counselling process is highly dependent on experienced professionals, making it difficult to provide individualized guidance to large student populations. Furthermore, recommendations often rely on subjective interpretation of psychometric assessments and interviews, leading to variations in recommendation quality and consistency [38]. The increasing number of students seeking academic and career guidance has therefore motivated researchers to investigate automated decision-support systems capable of delivering consistent recommendations at scale.

The widespread adoption of computers during the late twentieth century led to the development of Computer-Assisted Career Guidance Systems (CACGS), which represented one of the earliest attempts to automate vocational decision-making [18]. These systems typically employed predefined decision trees, expert knowledge bases, and rule-based inference mechanisms to recommend career options based on questionnaire responses. While Computer-Assisted Career Guidance Systems significantly improved accessibility and consistency compared with manual counselling, their decision-making capabilities remained constrained by manually encoded rules and predefined knowledge bases. Consequently, they lacked the flexibility required to interpret complex or ambiguous user inputs.

The emergence of Machine Learning introduced a significant shift from rule-based reasoning to data-driven decision-making. Instead of relying exclusively on handcrafted rules, supervised learning algorithms enabled systems to learn predictive patterns directly from labelled datasets [9]. Classification algorithms such as Logistic Regression, Support Vector Machines (SVM), Decision Trees, Random Forests, and Naïve Bayes demonstrated remarkable success in educational analytics, text classification, and recommendation systems due to their ability to generalize from historical data [13, 21, 20]. These techniques reduced dependence on manually designed rules while improving classification accuracy and adaptability.

Simultaneously, Natural Language Processing emerged as a key enabling technology for analyzing unstructured textual information. Earlier recommendation systems primarily relied on keyword matching and statistical representations such as Bag-of-Words (BoW) and Term Frequency–Inverse Document Frequency (TF-IDF) to process textual data [2, 1]. Although these methods remain computationally efficient and interpretable, they generally ignore contextual relationships between words and therefore struggle to capture semantic meaning. This limitation often results in inaccurate recommendations when users describe similar interests using different vocabulary or writing styles.

A major breakthrough occurred with the introduction of transformer-based language models, particularly Bidirectional Encoder Representations from Transformers (BERT), which enabled deep contextual language understanding through self-attention mechanisms [4]. Building upon this foundation, Sentence-BERT (SBERT) introduced a Siamese network architecture capable of generating semantically meaningful sentence embeddings suitable for efficient similarity computation [3]. These developments substantially improved the capability of recommendation systems to understand user intent expressed through natural language and have become fundamental components of modern intelligent advisory systems.

Recent research has further expanded career recommendation by integrating multiple Artificial Intelligence techniques within hybrid recommendation frameworks. Contemporary systems increasingly combine supervised classification, semantic similarity analysis, personality modelling, collaborative filtering, content-based recommendation, and explainable Artificial Intelligence to generate recommendations that are not only accurate but also transparent and personalized [14, 25, 39]. The integration of these

complementary techniques addresses many of the limitations associated with traditional counselling and earlier rule-based systems while improving recommendation quality, interpretability, and user trust.

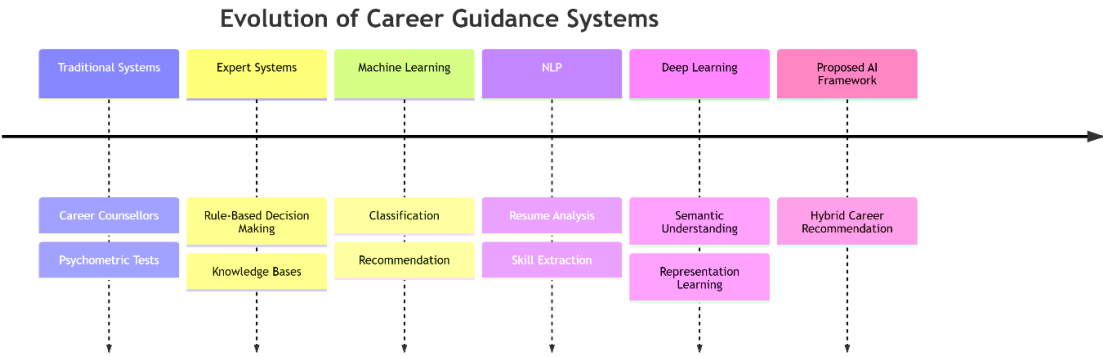


Figure 2.1 Stages in the Evolution of Career Guidance Systems: From Traditional Counselling to AI-Driven Hybrid Recommendation

Figure 2.1 illustrates the major stages in the evolution of career guidance systems, highlighting the transition from traditional human-centered counselling to intelligent hybrid recommendation frameworks. This evolution demonstrates a clear progression toward increasingly automated, context-aware, and personalized career guidance solutions. The proposed AI-driven career guidance system presented in this dissertation builds upon these advancements by integrating supervised Machine Learning, transformer-based semantic representations, personality-aware analysis, clustering, and hybrid recommendation techniques within a unified explainable framework.

Table 2.1 Comparative Overview of Career Guidance System Generations by Approach, Strengths, and Limitations

Generation	Primary Approach	Strengths	Limitations
Traditional Counselling	Interviews, Aptitude Tests, Psychometric Assessment	Personalized human interaction	Subjective, time-consuming, limited scalability
Computer-Assisted Systems	Rule-Based Expert Systems	Consistent recommendations	Static knowledge base, limited adaptability

Generation	Primary Approach	Strengths	Limitations
Machine Learning Systems	Supervised Classification Models	Learns from data, improved prediction accuracy	Limited semantic understanding
NLP-based Systems	Statistical Text Processing (TF-IDF, BoW)	Efficient text representation	Unable to capture contextual semantics
Transformer-based Systems	BERT, SBERT	Deep semantic understanding	Higher computational complexity
Hybrid AI Systems	ML + NLP + Semantic Embeddings + Personality Recommendation	Personalized, explainable, scalable recommendations	Increased architectural complexity

2.3 Traditional Career Counselling Approaches

Career counselling has traditionally been regarded as an essential component of educational and professional development, assisting individuals in identifying suitable career pathways based on their interests, abilities, personality traits, and aspirations [16, 23]. Over the past century, several theoretical models and psychological frameworks have been proposed to support career decision-making, many of which continue to influence modern counselling practices. Although these approaches provide valuable insights into human behavior and vocational preferences, their practical implementation often depends on expert interpretation and structured assessment methods, limiting their scalability in contemporary educational environments.

One of the earliest and most influential contributions to career guidance was made by Frank Parsons, who is widely recognized as the father of vocational guidance. In his seminal work *Choosing a Vocation* (1909), Parsons proposed that effective career selection should be based on three fundamental principles: a clear understanding of an individual's personal characteristics, comprehensive knowledge of occupational opportunities, and

logical reasoning to match personal attributes with appropriate careers [16]. This "trait-and-factor" approach established the foundation for scientific career counselling and inspired numerous vocational assessment methodologies developed in subsequent decades.

Building upon these early concepts, John L. Holland introduced the Theory of Vocational Personalities and Work Environments, commonly known as the RIASEC model, which remains one of the most widely adopted frameworks in career counselling [15]. Holland proposed that both individuals and occupations can be classified into six personality types: Realistic (R), Investigative (I), Artistic (A), Social (S), Enterprising (E), and Conventional (C). According to this theory, individuals experience greater job satisfaction, productivity, and career stability when their personality characteristics closely align with their occupational environment. Due to its simplicity and practical applicability, the RIASEC model has been extensively employed by educational institutions, career counselling centres, and vocational assessment platforms.

Another widely accepted psychological framework is the Five-Factor Model (FFM), commonly referred to as the Big Five Personality Traits, which evaluates human personality across five independent dimensions: Openness to Experience, Conscientiousness, Extraversion, Agreeableness, and Neuroticism [17, 35]. Unlike vocational interest models that primarily focus on occupational preferences, the Big Five framework provides a broader understanding of behavioral tendencies, interpersonal interactions, emotional stability, and cognitive characteristics. Numerous empirical studies have demonstrated significant relationships between personality traits and career preferences, leadership potential, workplace performance, and long-term professional satisfaction [27, 35]. Consequently, personality assessment has become an important component of modern career guidance systems.

Traditional career counselling also incorporates psychometric assessments, aptitude tests, interest inventories, and structured interviews to evaluate individual competencies and recommend suitable career pathways [38]. Psychometric instruments measure cognitive abilities, reasoning skills, numerical aptitude, verbal proficiency, and personality characteristics, while structured interviews allow counsellors to understand an individual's educational background, career aspirations, and motivational factors. These assessments provide valuable information for personalized counselling but require considerable human expertise and interpretation.

Despite their widespread adoption, conventional career counselling approaches exhibit several limitations that reduce their effectiveness in large-scale educational environments. First, the counselling process is highly dependent on experienced professionals, making it difficult to provide individualized guidance for institutions with large student populations [23]. Second, recommendations are often influenced by subjective judgement, leading to variations in decision quality among different counsellors. Third, most traditional approaches rely on predefined questionnaires that restrict users to fixed response options, thereby limiting their ability to express complex interests and aspirations through natural language [38]. Furthermore, these methods generally lack mechanisms for continuously adapting recommendations based on evolving industry trends, emerging technologies, or changing labor market requirements.

The rapid expansion of higher education, online learning platforms, and digital career services has further exposed the scalability limitations of manual counselling. Educational institutions increasingly require intelligent systems capable of providing personalized recommendations to thousands of students simultaneously while maintaining consistency and transparency [22]. These requirements have motivated researchers to investigate Artificial Intelligence-based career guidance systems capable of analyzing user behavior, processing free-form textual information, and generating personalized recommendations through computational models rather than manual interpretation.

Although traditional counselling frameworks continue to provide valuable theoretical foundations for understanding vocational interests and personality characteristics, they are insufficient to address the demands of modern educational ecosystems characterized by large-scale data, dynamic labor markets, and diverse learner profiles. Consequently, recent research has shifted towards integrating Artificial Intelligence, Natural Language Processing, Machine Learning, and recommender systems to overcome these limitations while preserving the psychological principles established by traditional career counselling theories.

The proposed career guidance framework presented in this dissertation builds upon these established psychological foundations while extending them through Artificial Intelligence techniques capable of processing natural language inputs, understanding semantic relationships, and generating explainable, personalized career recommendations. By integrating behavioral insights with data-driven computational models, the proposed

approach aims to provide scalable and context-aware career guidance suitable for modern educational environments.

Table 2.2 – Comparison of Traditional Career Counselling Frameworks

Framework	Primary Focus	Strengths	Limitations
Trait-and-Factor Theory (Frank Parsons)	Matching personal traits with occupations	Scientific foundation for career counselling	Manual interpretation and limited scalability
Holland's RIASEC Model	Personality–occupation alignment	Easy to understand and widely adopted	Requires structured questionnaires and expert interpretation
Big Five Personality Model	Behavioral and psychological characteristics	Comprehensive personality assessment	Does not directly recommend career paths
Psychometric & Aptitude Tests	Skills, interests, and cognitive abilities	Standardized evaluation	Time-consuming and dependent on expert administration
Traditional Counselling Interviews	Personal interaction and guidance	Highly personalized	Subjective, labor-intensive, and difficult to scale

2.4 Natural Language Processing for Career Guidance

Natural Language Processing (NLP) is a multidisciplinary field that combines Artificial Intelligence, Computational Linguistics, and Machine Learning to enable computers to understand, analyze, interpret, and generate human language [1, 9]. Over the past two decades, NLP has become one of the fundamental technologies supporting intelligent decision-support systems by enabling automated processing of large volumes of unstructured textual information. In educational applications, NLP has been successfully employed for intelligent tutoring systems, automated essay evaluation, sentiment analysis,

question-answering systems, recommendation systems, and career guidance platforms [43, 6].

Unlike structured databases, user-generated career descriptions are typically expressed as free-form text containing diverse vocabulary, grammatical variations, abbreviations, and contextual information. Consequently, an effective career recommendation system must first convert this unstructured textual information into a structured representation that can be processed by computational models. NLP provides a systematic pipeline for transforming raw text into meaningful features while preserving relevant semantic information [1, 1].

The first stage of the NLP pipeline involves text preprocessing, which aims to improve data quality by reducing linguistic variations and eliminating irrelevant information. Common preprocessing operations include text normalization, tokenization, stop-word removal, punctuation removal, case normalization, stemming, and lemmatization [9, 6]. Text normalization converts all characters into a consistent format, while tokenization divides sentences into individual words or tokens. Stop-word removal eliminates frequently occurring words that contribute little semantic information, whereas stemming and lemmatization reduce words to their root or base forms, thereby minimizing vocabulary size and improving model generalization.

These preprocessing techniques not only reduce computational complexity but also improve the quality of feature extraction by minimizing lexical redundancy. Numerous studies have demonstrated that appropriate preprocessing significantly improves classification accuracy, particularly for supervised machine learning models trained on textual datasets [6, 9].

After preprocessing, textual information must be transformed into numerical representations suitable for machine learning algorithms. One of the most widely adopted statistical text representation techniques is Term Frequency–Inverse Document Frequency (TF-IDF), originally proposed to measure the importance of a word within a document relative to a collection of documents [2, 1]. TF-IDF assigns higher weights to discriminative terms that occur frequently within a document but infrequently across the corpus, thereby emphasizing words that contribute significantly to document representation.

Due to its computational efficiency, simplicity, and interpretability, TF-IDF has become a standard feature extraction technique for document classification, information retrieval, spam detection, sentiment analysis, and educational text mining [2, 1, 30]. In career recommendation systems, TF-IDF enables effective representation of user interests by highlighting important career-related keywords while reducing the influence of commonly occurring words.

Despite these advantages, TF-IDF suffers from several inherent limitations. The technique assumes that words occurring independently represent document meaning and therefore ignores contextual relationships between neighbouring words. Consequently, semantically similar sentences containing different vocabulary often produce significantly different vector representations. Furthermore, TF-IDF cannot distinguish between synonyms, polysemous words, or contextual variations, limiting its ability to understand user intent expressed through natural language [4, 6].

To overcome these limitations, researchers introduced distributed word representations such as Word2Vec, GloVe, and FastText, which learn dense vector representations by analyzing word co-occurrence patterns within large textual corpora [6]. Unlike TF-IDF, these embedding techniques capture semantic relationships between words by placing semantically similar words closer together in the embedding space. For example, words such as *developer*, *programmer*, and *software engineer* exhibit similar vector representations despite being lexically different.

Although distributed word embeddings represented a significant improvement over statistical feature extraction methods, they generate a single embedding for each word irrespective of context. Consequently, words with multiple meanings receive identical vector representations regardless of their usage within different sentences. This limitation motivated the development of contextual language models based on transformer architectures, which dynamically generate context-dependent word representations [4].

Modern career recommendation systems increasingly employ transformer-based semantic models because they understand complete sentence meaning rather than relying solely on keyword frequency or isolated word embeddings. These contextual representations significantly improve semantic similarity computation, enabling recommendation systems

to recognize equivalent user intentions expressed using different vocabulary or sentence structures [3, 25].

Considering the computational efficiency of TF-IDF and the superior contextual understanding offered by transformer-based semantic models, recent research increasingly advocates hybrid approaches that combine traditional statistical feature extraction with contextual semantic representations [43]. Such approaches benefit from the interpretability and efficiency of classical NLP techniques while leveraging deep contextual understanding provided by modern transformer architectures.

Table 2.3 Natural Language Processing Pipeline for Career Recommendation

Technique	Representation	Advantages	Limitations	Typical Applications
Bag of Words	Frequency counts	Simple and efficient	Ignores word order and semantics	Text classification
TF-IDF	Weighted frequency	Interpretable and computationally efficient	No contextual understanding	Information retrieval, document classification
Word2Vec	Dense word embeddings	Captures semantic similarity	Static embeddings	NLP applications
GloVe	Global word embeddings	Utilizes global co-occurrence statistics	Static embeddings	Language modelling
FastText	Sub-word embeddings	Handles rare and unseen words	Context-independent	Text classification
BERT	Contextual embeddings	Deep contextual understanding	Computationally intensive	Question answering, semantic search

Technique	Representation	Advantages	Limitations	Typical Applications
Sentence-BERT	Sentence embeddings	Efficient semantic similarity computation	Higher computational cost than TF-IDF	Recommendation systems, semantic retrieval

In the proposed AI-driven career guidance framework, Natural Language Processing forms the foundational component responsible for analyzing user input before subsequent Machine Learning classification and semantic similarity computation. The combination of traditional preprocessing, TF-IDF feature extraction, and transformer-based semantic modelling enables the proposed system to effectively interpret free-form user descriptions while supporting accurate and personalized career recommendations.

2.5 Machine Learning Techniques for Text Classification

Machine Learning (ML) has emerged as one of the fundamental technologies for analyzing textual data and developing intelligent decision-support systems. Unlike traditional rule-based approaches, Machine Learning algorithms automatically learn discriminative patterns from labelled datasets, enabling them to classify previously unseen data with high accuracy [9]. In Natural Language Processing (NLP), supervised Machine Learning has been extensively employed for applications such as document classification, sentiment analysis, spam detection, topic identification, information retrieval, and educational analytics [6, 1]. These applications have demonstrated the capability of ML algorithms to transform high-dimensional textual representations into meaningful predictive models.

The success of supervised Machine Learning depends largely on the availability of labelled datasets and effective feature extraction techniques. Following text preprocessing and feature representation using methods such as TF-IDF, machine learning classifiers identify statistical relationships between input features and predefined output classes [2, 9]. In career recommendation systems, these classifiers play a crucial role in automatically predicting the most appropriate career domain based on users' free-form textual descriptions, thereby reducing the search space for subsequent recommendation modules.

Several supervised learning algorithms have been widely investigated for text classification tasks. Among the most commonly adopted techniques are Naïve Bayes, Support Vector Machines (SVM), Decision Trees, Random Forests, and Logistic Regression, each possessing distinct advantages and limitations depending on the characteristics of the dataset and application domain [8, 12, 13, 28].

Naïve Bayes is one of the earliest probabilistic classifiers used for document classification. It assumes conditional independence among features and estimates the probability of each class using Bayes' theorem [28]. Due to its simplicity, low computational complexity, and ability to handle high-dimensional feature spaces, Naïve Bayes has been widely employed in spam filtering, sentiment analysis, and document categorization. However, the independence assumption rarely holds for natural language data, resulting in reduced performance when complex contextual relationships exist among words.

Support Vector Machines (SVM) have demonstrated excellent performance in high-dimensional text classification problems by identifying an optimal separating hyperplane that maximizes the margin between different classes [13]. SVM classifiers are particularly effective when dealing with sparse textual representations such as TF-IDF vectors and often outperform simpler probabilistic models. Nevertheless, selecting appropriate kernel functions and hyperparameters can significantly influence classification performance, while computational complexity increases for large-scale datasets.

Decision Trees classify data by recursively partitioning the feature space according to information gain or impurity measures [27]. Their primary advantage lies in interpretability, as the resulting tree structure can be easily understood by human users. However, individual decision trees frequently suffer from overfitting and unstable decision boundaries when trained on noisy or high-dimensional textual datasets.

To overcome these limitations, Random Forests combine multiple decision trees through ensemble learning, thereby improving generalization capability and reducing overfitting [12]. Random Forest classifiers have demonstrated robust performance across numerous classification tasks by aggregating predictions from multiple independently trained trees. Despite their improved predictive capability, Random Forest models are computationally more expensive than linear classifiers and generally provide less interpretable decision boundaries.

Among these supervised learning techniques, Logistic Regression has consistently remained one of the most effective baseline algorithms for text classification using sparse feature representations such as TF-IDF [8]. Logistic Regression models the probability of class membership through a logistic function, producing probabilistic outputs that are both interpretable and computationally efficient. The algorithm performs particularly well for high-dimensional sparse feature spaces commonly encountered in Natural Language Processing applications because it learns linear decision boundaries while maintaining strong generalization performance [13, 15].

Several comparative studies have reported that Logistic Regression achieves competitive performance relative to more complex classification algorithms for document classification tasks, particularly when feature engineering techniques such as TF-IDF are employed [42]. Furthermore, the algorithm requires relatively few hyperparameters, exhibits stable convergence behavior, and provides interpretable feature weights, making it particularly attractive for educational decision-support systems where transparency and computational efficiency are important considerations.

Although deep learning approaches have recently demonstrated superior performance on several Natural Language Processing benchmarks, classical Machine Learning algorithms continue to offer significant practical advantages for medium-sized labelled datasets. These advantages include lower computational requirements, faster training times, easier interpretability, and reduced dependence on specialized hardware [44]. Consequently, many modern recommendation systems adopt hybrid architectures that combine traditional Machine Learning classifiers with transformer-based semantic embedding models to leverage the strengths of both approaches.

Table 2.4 Overview of Supervised Machine Learning Algorithms for Text Classification

Algorithm	Advantages	Limitations	Suitability for Career Guidance
Naïve Bayes	Fast training, simple implementation	Assumes feature independence	Moderate
Decision Tree	Highly interpretable	Overfitting on complex datasets	Moderate

Algorithm	Advantages	Limitations	Suitability for Career Guidance
Random Forest	High accuracy, robust	Computationally expensive	High
Support Vector Machine	Excellent for high-dimensional sparse data	Hyperparameter sensitive	High
Logistic Regression	Computationally efficient, interpretable, stable, probabilistic output	Linear decision boundary	Very High (Selected)

Considering these factors, the proposed career guidance system employs TF-IDF feature extraction together with Logistic Regression for career domain classification. This combination provides an effective balance between predictive accuracy, computational efficiency, interpretability, and implementation simplicity. The domain predictions generated by Logistic Regression subsequently guide the semantic recommendation stage, where transformer-based Sentence-BERT embeddings are employed to identify the most relevant career profiles. This modular architecture enables the proposed framework to combine the efficiency of classical Machine Learning with the contextual understanding provided by modern transformer models.

2.6 Transformer Models and Semantic Representation

The rapid growth of Natural Language Processing (NLP) has led to significant advancements in language representation techniques, progressing from statistical methods to deep contextual language models capable of understanding semantic relationships within text. Traditional text representation techniques such as Bag-of-Words (BoW) and TF-IDF primarily rely on word frequency statistics and therefore fail to capture contextual meaning or semantic relationships between words [2, 1]. Although these methods remain computationally efficient and interpretable, they often generate different representations for semantically equivalent sentences containing different vocabulary, thereby limiting their effectiveness in intelligent recommendation systems.

To address these limitations, researchers introduced distributed word representation techniques such as Word2Vec, GloVe, and FastText, which learn dense vector representations based on word co-occurrence patterns within large textual corpora [6]. These embedding models significantly improved semantic representation by placing semantically related words closer together in the embedding space. For example, words such as *developer*, *programmer*, and *software engineer* are represented by similar vectors despite having different lexical forms. However, these approaches generate a single vector representation for each word regardless of its surrounding context. Consequently, words with multiple meanings receive identical embeddings even when their contextual interpretations differ.

A major breakthrough in Natural Language Processing occurred with the introduction of the Transformer architecture by Vaswani et al. in 2017 [26]. Unlike recurrent neural networks (RNNs) and long short-term memory (LSTM) models, transformers employ a self-attention mechanism that enables simultaneous processing of all words within a sentence while modelling long-range dependencies more effectively. The self-attention mechanism assigns different importance weights to words depending on their contextual relationships, allowing the model to generate richer and more meaningful language representations. This architecture substantially improved computational efficiency through parallel processing while achieving state-of-the-art performance across multiple Natural Language Processing benchmarks.

Building upon the transformer architecture, Bidirectional Encoder Representations from Transformers (BERT) was introduced by Devlin et al. in 2019 as a pre-trained language representation model that learns bidirectional contextual information using masked language modelling and next sentence prediction tasks [4]. Unlike previous embedding models, BERT generates context-dependent representations for each word by considering both preceding and succeeding words within a sentence. Consequently, identical words may receive different embeddings depending on their contextual usage, enabling superior understanding of ambiguity, syntax, and semantics.

Although BERT demonstrated remarkable performance across numerous Natural Language Processing tasks, its application to semantic similarity and information retrieval [31] remained computationally expensive. Semantic comparison using standard BERT requires evaluating each sentence pair independently, resulting in substantial

computational overhead for large-scale recommendation systems [4]. This limitation significantly restricts its practical deployment in applications requiring real-time similarity computation among thousands of candidate documents.

To overcome this limitation, Reimers and Gurevych proposed Sentence-BERT (SBERT), which extends the BERT architecture using a Siamese neural network structure capable of generating fixed-length sentence embeddings [3]. Instead of comparing sentence pairs individually, SBERT independently encodes each sentence into a dense vector representation. Semantic similarity can subsequently be computed efficiently using cosine similarity between embedding vectors, making SBERT highly suitable for recommendation systems, semantic retrieval, duplicate question detection, clustering, and information retrieval tasks.

Compared with conventional BERT, Sentence-BERT substantially reduces computational complexity while preserving contextual semantic understanding. Numerous experimental studies have demonstrated that SBERT achieves superior performance on Semantic Textual Similarity (STS) benchmarks and recommendation tasks while reducing inference time by several orders of magnitude [3]. These characteristics make SBERT particularly appropriate for intelligent career recommendation systems, where semantic comparison must be performed between user interests and hundreds of career descriptions in real time.

Several recent studies have successfully applied transformer-based semantic models within educational recommendation systems, demonstrating improvements in recommendation relevance, semantic retrieval accuracy, and personalization compared with traditional keyword-based approaches [50, 51]. These studies indicate that contextual semantic understanding enables recommendation systems to identify equivalent meanings expressed through different vocabulary, thereby reducing dependence on exact keyword matching.

Despite these advantages, transformer models also present certain challenges. Training and fine-tuning transformer architectures require considerable computational resources, large-scale datasets, and specialized hardware such as Graphics Processing Units (GPUs). Furthermore, transformer models typically exhibit higher memory consumption and inference complexity compared with traditional statistical approaches [46]. Consequently, many practical applications adopt hybrid architectures that combine lightweight machine

learning classifiers for coarse-grained prediction with transformer-based semantic models for fine-grained recommendation.

Table 2.5 Evolution of Language Representation Techniques

Model	Context Awareness	Semantic Understanding	Computational Cost	Suitable for Recommendation
Bag of Words	✗	Low	Very Low	Low
TF-IDF	✗	Low	Low	Moderate
Word2Vec	Partial	Moderate	Moderate	Moderate
GloVe	Partial	Moderate	Moderate	Moderate
FastText	Partial	Moderate	Moderate	High
BERT	✓	Very High	Very High	High
Sentence-BERT	✓	Very High	Moderate	Very High (Selected)

Considering these observations, the proposed career guidance framework integrates Sentence-BERT as the semantic similarity component rather than employing it as a standalone classifier. Initially, TF-IDF and Logistic Regression are used to identify the most relevant career domain, thereby reducing the search space. Subsequently, Sentence-BERT computes semantic similarity between the user's interests and career profiles within the predicted domain. This modular architecture combines the computational efficiency of classical Machine Learning with the contextual understanding of transformer models, enabling accurate, scalable, and explainable career recommendations.

2.7 Personality Modelling in AI-based Career Recommendation

Personality plays a significant role in influencing an individual's career interests, workplace behavior, learning preferences, and long-term professional satisfaction. While technical skills and educational qualifications determine an individual's capability to perform a particular job, personality characteristics often influence how effectively an individual adapts to different work environments, collaborates with others, and responds to professional challenges [17, 35]. Consequently, personality assessment has become an

important component of modern career guidance systems, enabling recommendations that consider not only skills and interests but also behavioral tendencies and personal preferences.

Early career guidance systems primarily focused on aptitude assessments and vocational interest inventories, often overlooking behavioral characteristics that influence career success. Psychological research has demonstrated that individuals possessing similar technical abilities may perform differently in the same profession because of differences in motivation, emotional stability, interpersonal communication, and adaptability [44, 27]. These findings have motivated researchers to integrate psychological theories into computational recommendation systems, thereby improving recommendation relevance and user satisfaction.

Among the various personality assessment frameworks proposed in psychology, the Five-Factor Model (FFM), commonly known as the Big Five Personality Traits, is the most widely accepted and empirically validated model [17, 35]. The framework describes personality using five independent dimensions:

- Openness to Experience – creativity, curiosity, innovation, and willingness to explore new ideas.
- Conscientiousness – organization, discipline, responsibility, and goal-oriented behavior.
- Extraversion – sociability, communication skills, leadership, and enthusiasm.
- Agreeableness – cooperation, empathy, trust, and teamwork.
- Neuroticism – emotional stability, stress management, and resilience.

Research has consistently demonstrated significant relationships between these personality dimensions and occupational preferences. Individuals exhibiting high openness are often attracted to research, design, and creative professions, whereas highly conscientious individuals generally perform well in structured and detail-oriented occupations such as engineering, finance, and project management [17, 34]. Similarly, extraversion is positively associated with leadership, management, sales, and public-facing professions, while agreeableness contributes to success in healthcare, education, counselling, and social services. Emotional stability influences an individual's ability to perform effectively under pressure, making it relevant across diverse professional domains.

Traditional personality assessment typically relies on structured psychometric questionnaires consisting of multiple-choice or Likert-scale responses. Although these instruments provide reliable personality measurements, they require users to complete lengthy assessments, which may reduce user engagement and limit practical adoption in large-scale educational environments [17]. Moreover, questionnaire-based assessments capture personality at a single point in time and may not fully reflect changes in behavior, interests, or professional aspirations over time.

Recent advances in Natural Language Processing have enabled researchers to infer personality characteristics directly from textual information by analyzing linguistic patterns, writing styles, vocabulary usage, and semantic structures [35, 35]. Machine Learning and transformer-based language models have demonstrated promising performance in personality prediction by identifying behavioral cues embedded within free-form textual expressions. Such approaches reduce dependence on lengthy questionnaires while enabling seamless integration of personality modelling into intelligent recommendation systems.

Several AI-driven recommendation systems have incorporated personality information to improve personalization and recommendation relevance. In educational recommendation systems, personality-aware models have been shown to improve learner engagement, recommendation diversity, and long-term user satisfaction compared with conventional content-based approaches [24]. Similarly, career recommendation systems integrating behavioral characteristics have demonstrated better alignment between recommended career paths and individual preferences, thereby increasing user confidence in automated recommendations.

Despite these advantages, personality modelling presents several practical challenges. Personality traits are inherently complex and dynamic, making accurate prediction difficult without sufficient behavioral evidence. Furthermore, inferred personality characteristics should complement rather than replace explicit user interests and educational qualifications [35]. Ethical considerations, including user privacy, informed consent, and responsible use of behavioral information, are also important when integrating personality analysis into Artificial Intelligence systems.

Considering these factors, the proposed career guidance framework incorporates personality-aware analysis as one component of a broader hybrid recommendation strategy. Rather than performing full psychometric evaluation, the system derives behavioral indicators from user-provided interests and combines them with semantic similarity, keyword matching, and domain classification scores. This approach improves personalization while maintaining computational efficiency, transparency, and user interpretability. Consequently, personality modelling acts as a complementary decision-support mechanism rather than the sole basis for career recommendation.

2.8 Recommender Systems and Hybrid Recommendation Approaches

Recommender Systems (RS) are intelligent decision-support systems designed to assist users in discovering relevant items from a large collection of available alternatives based on their preferences, behavior, or contextual information [10, 25]. Originally developed for e-commerce and digital content platforms, recommender systems have now become an essential component of numerous application domains, including healthcare, education, entertainment, finance, tourism, and career guidance. Their primary objective is to reduce information overload by automatically identifying and recommending the most relevant options that align with users' requirements and preferences.

In educational environments, recommender systems have been increasingly employed to support personalized learning, course selection, academic advising, skill development, and career planning [22]. As educational resources and career opportunities continue to expand rapidly, students often face difficulty identifying suitable academic pathways and professional opportunities. Intelligent recommendation systems address this challenge by analyzing user characteristics and recommending relevant career domains, learning resources, or employment opportunities.

Traditional recommender systems are generally classified into Content-Based Filtering, Collaborative Filtering, Knowledge-Based Recommendation, and Hybrid Recommendation Systems, each utilizing different sources of information for recommendation generation [10].

Content-Based Filtering (CBF) generates recommendations by comparing item attributes with user profiles or previously expressed interests [24]. In career recommendation

systems, content-based approaches analyze skills, educational qualifications, keywords, interests, and career descriptions to recommend occupations that closely resemble user preferences. Since recommendations are generated directly from item characteristics, content-based filtering does not require interaction data from other users. However, its effectiveness depends heavily on accurate feature representation and often suffers from limited recommendation diversity because users are typically recommended careers similar to those they have already expressed interest in.

Collaborative Filtering (CF) identifies patterns among users with similar preferences and recommends items preferred by like-minded individuals [11]. Collaborative filtering has achieved remarkable success in domains such as movie recommendation, online shopping, and music streaming by exploiting user-item interaction matrices. Nevertheless, its applicability to career recommendation remains limited because many educational environments lack sufficient historical interaction data. Furthermore, collaborative filtering suffers from the well-known cold-start problem, where recommendations for new users or newly introduced career profiles become difficult due to insufficient interaction history.

Knowledge-Based Recommendation Systems utilize explicit domain knowledge, predefined rules, ontologies, or expert-defined constraints to generate recommendations [32]. In career counselling applications, knowledge-based approaches often incorporate expert knowledge regarding educational qualifications, required technical skills, certification pathways, and occupational requirements. These systems provide highly interpretable recommendations but require substantial manual knowledge engineering and continuous updates to remain consistent with evolving labor market trends.

To overcome the limitations of individual recommendation strategies, researchers introduced Hybrid Recommendation Systems, which combine multiple recommendation techniques within a unified framework [25]. Hybrid recommender systems integrate complementary information sources such as user preferences, semantic similarity, behavioral patterns, collaborative interactions, personality characteristics, and domain knowledge to improve recommendation quality. By leveraging the strengths of different recommendation paradigms, hybrid systems generally achieve superior recommendation accuracy, diversity, robustness, and user satisfaction compared with individual recommendation approaches.

Several hybrid recommendation architectures have been proposed in the literature. Weighted hybrid methods combine prediction scores generated by multiple recommendation models using weighted aggregation. Switching hybrids dynamically select the most appropriate recommendation strategy depending on available information. Cascade hybrids apply recommendation algorithms sequentially, allowing one model to refine the recommendations generated by another. Feature augmentation approaches employ the output of one recommender as input features for another recommendation model [25]. The selection of a particular hybrid architecture depends upon application requirements, available datasets, computational complexity, and explainability considerations.

Table 2.6 Comparative Analysis of Recommendation Approaches

Recommendation Technique	Principle	Advantages	Limitations	Suitability for Career Guidance
Content-Based Filtering	Matches user profile with item attributes	Personalized, independent of other users	Limited diversity	High
Collaborative Filtering	Uses behavior of similar users	High recommendation accuracy	Cold-start problem, requires interaction history	Moderate
Knowledge-Based Recommendation	Uses expert knowledge and predefined rules	Explainable, domain-specific	Manual knowledge engineering	High
Hybrid Recommendation	Combines multiple recommendation strategies	High accuracy, diversity, explainability	Increased architectural complexity	Very High (Selected)

Recent advances in Artificial Intelligence have further expanded hybrid recommendation systems by integrating Machine Learning, Natural Language Processing, semantic

embeddings, behavioral modelling, and Explainable Artificial Intelligence (XAI) [14, 53]. In educational recommendation systems, these hybrid architectures have demonstrated improved recommendation relevance by combining multiple complementary perspectives rather than relying exclusively on keyword matching or collaborative behavior [40]. The integration of contextual semantic representations enables systems to understand user intent more accurately, while personality-aware analysis improves long-term recommendation suitability.

Despite these advancements, hybrid recommendation systems also present several challenges. Combining multiple Artificial Intelligence techniques increases architectural complexity, computational requirements, and parameter optimization efforts. Moreover, balancing the contribution of different recommendation components requires careful calibration to avoid introducing bias toward any individual recommendation strategy [46, 24]. Explainability also becomes increasingly important because users must understand how multiple recommendation signals contribute to the final recommendation score.

Considering these observations, the proposed AI-driven career guidance framework adopts a hybrid recommendation architecture that combines career domain prediction using Logistic Regression, semantic similarity using Sentence-BERT embeddings, keyword overlap analysis, personality-aware scoring, and rule-based weighting mechanisms. Unlike conventional recommendation systems that rely on a single recommendation strategy, the proposed framework integrates complementary Artificial Intelligence techniques to improve personalization, transparency, and recommendation quality. This design enables the system to exploit the strengths of both statistical Machine Learning and transformer-based semantic modelling while maintaining explainability and computational efficiency.

2.9 Clustering and Unsupervised Learning for Interest Grouping

Unlike supervised Machine Learning algorithms that require labelled datasets for training, unsupervised learning aims to discover hidden structures, relationships, and patterns within unlabeled data [9]. Among various unsupervised learning techniques, clustering is one of the most widely adopted approaches for exploratory data analysis because it groups similar observations based on their intrinsic characteristics without requiring predefined class labels. Clustering has found extensive applications in customer segmentation, healthcare

analytics, image processing, social network analysis, educational data mining, and recommender systems [7, 52].

In educational environments, clustering techniques are frequently employed to analyze student learning behavior, identify groups of learners with similar characteristics, evaluate academic performance, and discover hidden behavioral patterns [52]. These insights enable educational institutions to design personalized interventions, adaptive learning strategies, and recommendation systems that better address the needs of diverse learner populations. Similarly, career guidance systems utilize clustering to identify groups of users exhibiting similar interests, skills, or career aspirations, thereby supporting more effective recommendation generation and exploratory analysis.

Among the various clustering algorithms, K-Means clustering, proposed by MacQueen in 1967, remains one of the most widely used partition-based clustering techniques because of its conceptual simplicity, computational efficiency, and scalability [7]. The algorithm partitions a dataset into K disjoint clusters by iteratively assigning observations to the nearest cluster centroid while minimizing the within-cluster sum of squared distances. The iterative optimization process continues until cluster assignments stabilize or a predefined convergence criterion is satisfied.

The effectiveness of K-Means largely depends on the choice of the parameter K , which represents the number of clusters to be generated. Several techniques have been proposed to determine an appropriate value of K , including the Elbow Method, Silhouette Analysis, and the Gap Statistic [29, 48]. Among these approaches, the Elbow Method is widely adopted because it provides a simple visual interpretation by identifying the point at which increasing the number of clusters produces diminishing improvements in clustering performance.

K-Means has been successfully applied to textual datasets represented using TF-IDF vectors, Word Embeddings, and Sentence Embeddings, enabling the discovery of semantically related document groups [2, 3]. When combined with contextual semantic representations such as Sentence-BERT, clustering algorithms become capable of identifying groups of users expressing similar interests even when different vocabulary or sentence structures are used. This characteristic makes clustering particularly valuable in

career recommendation systems, where users often describe similar professional aspirations using diverse linguistic expressions.

One challenge associated with clustering high-dimensional textual data is visualization. Since textual feature representations often contain hundreds or thousands of dimensions, direct visualization becomes impractical. Consequently, dimensionality reduction techniques such as Principal Component Analysis (PCA) and t-distributed Stochastic Neighbor Embedding (t-SNE) are commonly employed to project high-dimensional feature spaces into two- or three-dimensional representations suitable for visualization [20, 47]. These visualization techniques enable researchers to inspect cluster separation, identify outliers, and evaluate clustering quality more effectively.

Several recent studies have demonstrated the effectiveness of clustering in educational recommendation systems by identifying learner profiles, grouping students according to behavioral characteristics, and discovering hidden educational patterns [49]. Similarly, clustering has been applied to career recommendation systems to categorize occupations, identify related career pathways, and improve recommendation diversity by exposing users to multiple career groups rather than limiting recommendations to a single predefined category [41].

Despite its widespread adoption, K-Means exhibits several limitations. The algorithm assumes that clusters are approximately spherical and of similar size, making it less effective for irregularly shaped data distributions [7]. Furthermore, the algorithm is sensitive to the initial placement of cluster centroids and may converge to local optima. The requirement to specify the number of clusters in advance also introduces additional complexity when analyzing previously unseen datasets. Nevertheless, due to its computational efficiency, scalability, and ease of interpretation, K-Means remains one of the most practical clustering algorithms for exploratory analysis of textual datasets.

Table 2.7: Comparison of Clustering Algorithms

Algorithm	Advantages	Limitations	Suitability for Career Recommendation
K-Means	Fast, scalable, easy to interpret	Requires predefined K, sensitive to initialization	Very High (Selected)
Hierarchical Clustering	No need to specify K initially	High computational complexity	Moderate
DBSCAN	Detects arbitrary-shaped clusters and noise	Sensitive to parameter selection	Moderate
Gaussian Mixture Models (GMM)	Soft clustering with probabilistic assignments	Computationally expensive	Moderate
Spectral Clustering	Effective for complex cluster structures	Poor scalability for large datasets	Low–Moderate

Considering these characteristics, the proposed AI-driven career guidance framework employs K-Means clustering as an exploratory analysis component rather than a primary recommendation algorithm. After career domain prediction and semantic similarity computation, clustering is applied to group users with similar interest patterns and visualize the distribution of career preferences. This supplementary analysis improves system interpretability by enabling users and administrators to understand broader trends within the recommendation space while supporting future enhancements based on behavioral clustering.

2.10 Existing AI-based Career Guidance Systems

Having reviewed the individual techniques in isolation, this section turns to how they have been combined in practice. Researchers have proposed numerous approaches for assisting students and job seekers in identifying suitable career paths using computational

techniques. These systems differ considerably in terms of the algorithms employed, the types of input data considered, the recommendation methodology adopted, and the degree of personalization provided [38, 39].

Early AI-based career guidance systems primarily relied on rule-based expert systems, where domain experts manually defined decision rules based on educational qualifications, aptitude scores, and occupational requirements [33]. Although these systems provided transparent and interpretable recommendations, their effectiveness depended entirely on manually constructed knowledge bases. Updating such systems to accommodate emerging technologies, evolving job roles, and changing labor market requirements proved challenging, limiting their long-term applicability.

As Machine Learning techniques matured, researchers shifted toward supervised classification models capable of automatically learning relationships between user characteristics and career domains. Algorithms such as Decision Trees, Support Vector Machines, Random Forests, Artificial Neural Networks, and Logistic Regression were successfully employed to predict suitable career categories using academic records, aptitude scores, personality traits, and textual information [42, 44]. These approaches demonstrated improved prediction accuracy compared with rule-based systems; however, they generally relied on structured input data and exhibited limited capability to interpret complex natural language descriptions.

To overcome the limitations of structured questionnaires, several researchers introduced Natural Language Processing-based career recommendation systems capable of analyzing free-form textual inputs [6]. Classical NLP approaches primarily employed TF-IDF vectorization, keyword matching, cosine similarity, and information retrieval techniques to compare user interests with predefined career descriptions. Although these systems significantly improved user flexibility by allowing open-ended textual input, they remained dependent on lexical similarity and often failed to recognize semantically equivalent expressions containing different vocabulary.

Recent advances in transformer-based language models have substantially improved semantic understanding within career recommendation systems. Several contemporary studies have employed contextual language models such as BERT, RoBERTa, and Sentence-BERT to analyze user interests, compute semantic similarity, and improve

recommendation relevance [4, 3, 50]. Compared with traditional keyword-based approaches, transformer-based systems demonstrate superior capability in understanding contextual relationships, identifying implicit user intentions, and generating semantically meaningful recommendations.

Another important research direction involves integrating psychological and behavioral characteristics into career recommendation systems. Several studies have combined personality assessment frameworks such as the Big Five Personality Traits with recommendation algorithms to improve recommendation relevance and long-term career satisfaction [34, 45]. Personality-aware recommendation systems provide recommendations that better align with behavioral preferences; however, many existing approaches require lengthy psychometric questionnaires, thereby reducing usability and user engagement.

Researchers have also explored hybrid recommendation frameworks that combine multiple Artificial Intelligence techniques within a single recommendation pipeline. Hybrid systems typically integrate supervised Machine Learning, semantic similarity analysis, collaborative filtering, content-based recommendation, behavioral modelling, and expert-defined rules to improve recommendation quality [25, 53]. These systems generally outperform single-method recommendation approaches because they exploit complementary information sources while reducing the weaknesses associated with individual algorithms.

Despite these advancements, a comprehensive review of recent literature reveals several persistent limitations. Many existing career recommendation systems focus primarily on either structured questionnaire responses or semantic similarity analysis without integrating multiple complementary Artificial Intelligence techniques. Similarly, several Machine Learning-based approaches achieve satisfactory classification performance but provide limited explanation regarding the rationale behind their recommendations. In addition, relatively few studies combine supervised classification, contextual semantic modelling, personality-aware analysis, clustering, and explainable recommendation mechanisms within a unified framework [39].

Table 2.8 Comparative Analysis of Existing AI-based Career Guidance Systems

Author	Year	Methodology	Dataset	Strengths	Limitations
Study 1	2021	Rule-based Recommendation	Student Survey	Explainable	Limited scalability
Study 2	2022	Machine Learning Classification	Academic Dataset	Good prediction accuracy	Structured input only
Study 3	2022	TF-IDF + Cosine Similarity	Career Descriptions	Simple implementation	Poor semantic understanding
Study 4	2023	BERT-based Recommendation	Text Corpus	Context-aware recommendation	Computationally expensive
Study 5	2023	Personality-aware Recommendation	Student Profiles	Better personalization	Questionnaire dependent
Study 6	2024	Hybrid Recommendation	Educational Dataset	Improved recommendation quality	Limited explainability
Proposed Framework	2026	TF-IDF + Logistic Regression + SBERT + Personality + K-Means + Hybrid Scoring	Free-form Career Interest Dataset	Semantic understanding, explainability, scalability, personalization	Requires model training and computational resources

Furthermore, most existing systems evaluate recommendation quality using classification accuracy alone while overlooking broader evaluation criteria such as semantic relevance, recommendation diversity, explainability, and user interpretability. These limitations

reduce the practical applicability of many proposed systems in real educational environments, where recommendations must be accurate, transparent, scalable, and adaptable to evolving labor market requirements.

The literature therefore indicates a clear need for an integrated Artificial Intelligence-driven career guidance framework capable of combining the strengths of multiple computational techniques while minimizing their individual limitations. The proposed framework presented in this dissertation addresses these challenges by integrating Natural Language Processing, TF-IDF-based domain classification, Sentence-BERT semantic similarity, personality-aware recommendation, K-Means clustering, and explainable hybrid scoring within a single unified recommendation architecture.

2.11 Research Gap Analysis

The comprehensive review of existing literature reveals that considerable progress has been achieved in the development of Artificial Intelligence-based career guidance systems. Researchers have successfully applied Machine Learning, Natural Language Processing, recommender systems, personality modelling, and transformer-based semantic representation techniques to improve recommendation accuracy and personalization [38, 39]. However, despite these advancements, several research challenges remain unresolved, limiting the effectiveness of existing career recommendation frameworks.

One of the most significant limitations observed in the literature is the dependence on structured questionnaires. Many existing career guidance systems require users to complete predefined psychometric assessments, aptitude tests, or multiple-choice questionnaires before recommendations can be generated [25, 27]. Although these approaches provide valuable behavioral information, they restrict users' ability to express their interests naturally and often reduce user engagement due to lengthy assessment procedures.

Another major limitation is the restricted semantic understanding exhibited by traditional Natural Language Processing approaches. Several career recommendation systems continue to rely on keyword matching, Bag-of-Words representations, TF-IDF vectors, or cosine similarity measures for comparing user interests with career descriptions [2, 1]. While these techniques are computationally efficient, they frequently fail to recognize

semantically equivalent expressions written using different vocabulary or sentence structures, thereby reducing recommendation relevance.

The literature also indicates that many Machine Learning-based recommendation systems primarily focus on classification accuracy while neglecting contextual semantic understanding [42]. Although supervised classifiers effectively predict career domains, they are generally incapable of capturing deeper semantic relationships between user interests and occupational descriptions. Consequently, relying solely on classification models often limits recommendation quality.

Transformer-based recommendation systems have substantially improved semantic understanding through contextual language representations [4, 3]. However, many existing studies employ transformer models as standalone recommendation engines without integrating complementary Artificial Intelligence techniques such as supervised classification, personality-aware analysis, or clustering. This increases computational complexity while limiting overall recommendation explainability and flexibility.

The review further reveals that personality modelling remains underutilized in many existing career recommendation systems. Several studies focus exclusively on technical skills, educational qualifications, or textual similarity without considering behavioral characteristics that influence long-term career satisfaction and workplace compatibility [34, 45]. Even among personality-aware systems, most require lengthy psychometric questionnaires instead of utilizing lightweight behavioral indicators derived from user interests.

Similarly, unsupervised learning techniques receive comparatively limited attention within existing career recommendation research. Although clustering algorithms have been widely employed in educational data mining and learner analytics, relatively few studies integrate clustering into career recommendation systems to analyze user interest patterns, improve recommendation interpretability, or support exploratory visualization [52, 41].

Table 2.9 Research Gap Analysis

Research Area	Existing Approaches	Identified Gap	Proposed Solution
User Input	Structured questionnaires	Limited flexibility for expressing interests	Free-form natural language input
Text Understanding	TF-IDF, keyword matching	Poor contextual semantic understanding	Sentence-BERT semantic embeddings
Domain Prediction	Standalone ML classifiers	Limited semantic reasoning	TF-IDF + Logistic Regression + SBERT
Personality Analysis	Questionnaire-based assessment	Time-consuming and low user engagement	Lightweight personality-aware scoring
Recommendation Strategy	Single recommendation technique	Limited personalization	Hybrid recommendation framework
Interest Analysis	Limited use of clustering	Poor interpretability	K-Means clustering with visualization
Explainability	Black-box recommendations	Low user trust	Transparent hybrid scoring mechanism

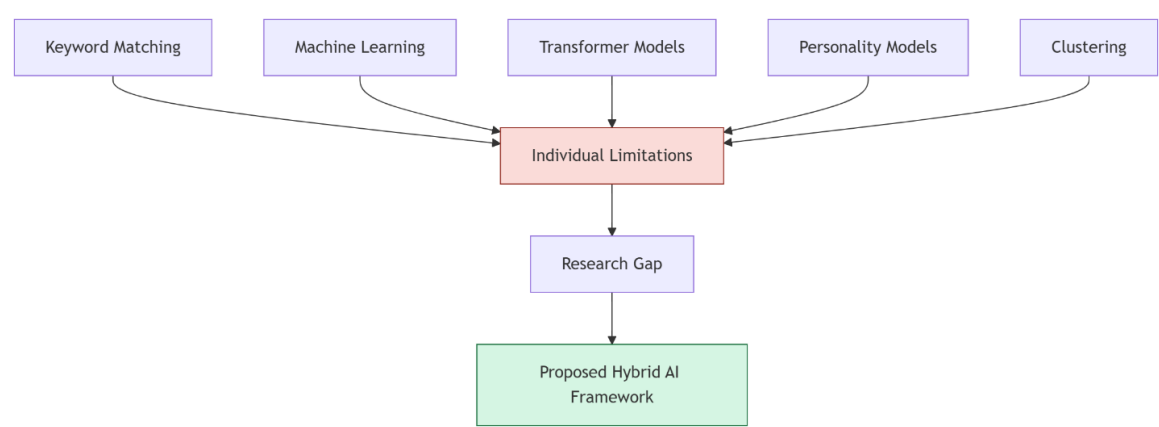


Figure 2.2 Research Gap Leading to the Proposed Framework

Another important observation concerns recommendation explainability. Numerous Artificial Intelligence-based recommendation systems prioritize predictive accuracy while providing little information regarding the reasoning behind generated recommendations [14]. Since career recommendations directly influence educational and professional decision-making, explainability is essential for improving user trust, transparency, and acceptance of automated decision-support systems.

Finally, the literature indicates that relatively few studies integrate Natural Language Processing, supervised Machine Learning, transformer-based semantic similarity, personality-aware analysis, clustering, and explainable hybrid recommendation within a single unified architecture. Most existing systems emphasize only one or two computational techniques, resulting in limitations regarding personalization, scalability, semantic understanding, or interpretability.

Based on the identified limitations, there exists a clear opportunity to develop an integrated Artificial Intelligence-driven career guidance framework capable of combining complementary computational techniques while maintaining computational efficiency, semantic understanding, recommendation explainability, and practical deployability. The proposed framework presented in this dissertation addresses these research gaps by integrating TF-IDF-based feature extraction, Logistic Regression-based career domain classification, Sentence-BERT semantic similarity, personality-aware scoring, K-Means clustering, and explainable hybrid recommendation mechanisms within a modular decision-support architecture. By combining these techniques, the proposed system aims to deliver more accurate, personalized, transparent, and scalable career recommendations compared with existing approaches.

2.12 Summary

This chapter presented a comprehensive review of the literature related to Artificial Intelligence-driven career guidance systems and the computational techniques that form the foundation of the proposed recommendation framework. The review began by examining the evolution of career guidance systems, highlighting the transition from

traditional counsellor-based approaches to intelligent decision-support systems powered by Artificial Intelligence, Machine Learning, and Natural Language Processing.

The chapter subsequently reviewed conventional career counselling theories, including the Trait-and-Factor approach, Holland's RIASEC model, and the Big Five Personality Traits, demonstrating their continued relevance in understanding vocational interests and behavioral characteristics. Although these psychological frameworks provide valuable theoretical foundations, the review identified several practical limitations, including dependence on expert interpretation, limited scalability, and reliance on structured questionnaires.

A detailed review of Natural Language Processing techniques illustrated the progression from statistical text representation methods such as TF-IDF to modern contextual language models based on transformer architectures. Similarly, the review of supervised Machine Learning techniques demonstrated the suitability of algorithms such as Logistic Regression for career domain classification using high-dimensional textual representations. The discussion further highlighted the advantages of Sentence-BERT in capturing contextual semantic relationships between user interests and career descriptions, thereby improving recommendation relevance beyond traditional keyword-based approaches.

The literature review also examined personality-aware recommendation systems, recommender system architectures, and unsupervised learning techniques such as K-Means clustering. These studies demonstrated that integrating behavioral information, semantic understanding, and exploratory data analysis significantly enhances recommendation personalization, interpretability, and user satisfaction. Furthermore, hybrid recommendation frameworks were found to outperform individual recommendation techniques by combining complementary computational methods while mitigating the limitations associated with standalone approaches.

The review of existing AI-based career guidance systems revealed that although considerable progress has been achieved in applying Artificial Intelligence to career recommendation, several research challenges remain unresolved. Existing systems frequently depend on structured questionnaires, exhibit limited contextual understanding, underutilize personality modelling and clustering techniques, and provide insufficient explanation regarding the reasoning behind generated recommendations. Moreover,

relatively few studies integrate supervised Machine Learning, transformer-based semantic similarity, behavioral analysis, clustering, and explainable recommendation strategies within a unified framework.

Based on the identified research gaps, this dissertation proposes a hybrid Artificial Intelligence-driven career guidance framework that combines TF-IDF-based feature extraction, Logistic Regression for career domain classification, Sentence-BERT semantic similarity, personality-aware analysis, K-Means clustering, and explainable hybrid recommendation techniques. This integrated approach aims to provide accurate, personalized, scalable, and transparent career recommendations while addressing the major limitations identified in existing literature.

The insights obtained from this literature review provide the theoretical and methodological foundation for the proposed research. The next chapter presents the overall system architecture, methodology, algorithms, implementation workflow, and design decisions adopted for developing the proposed AI-driven career guidance system.

CHAPTER 3: SYSTEM DESIGN AND METHODOLOGY

3.1 Introduction

Chapter 2 identified a specific research gap: existing career-guidance systems tend to apply natural language processing, semantic modelling, personality inference, and clustering in isolation, rather than integrating them into a single explainable framework. This chapter presents the design adopted to close that gap — the system architecture, requirement specification, datasets, and the methodology of each computational module (semantic similarity, clustering, personality inference, and hybrid scoring) that together transform a free-form user statement into a ranked, explainable set of career recommendations. Chapters 4 and 5 then report how this design was implemented and what its evaluation showed.

3.2 Overall System Architecture

The system adopts a modular, layered architecture rather than a monolithic implementation, so that individual components (preprocessing, classification, semantic similarity, personality inference, clustering, hybrid scoring, presentation) can be developed, tested, and upgraded independently. Figure 3.1 illustrates the six functional layers, and Table 3.1 summarizes the role of each.

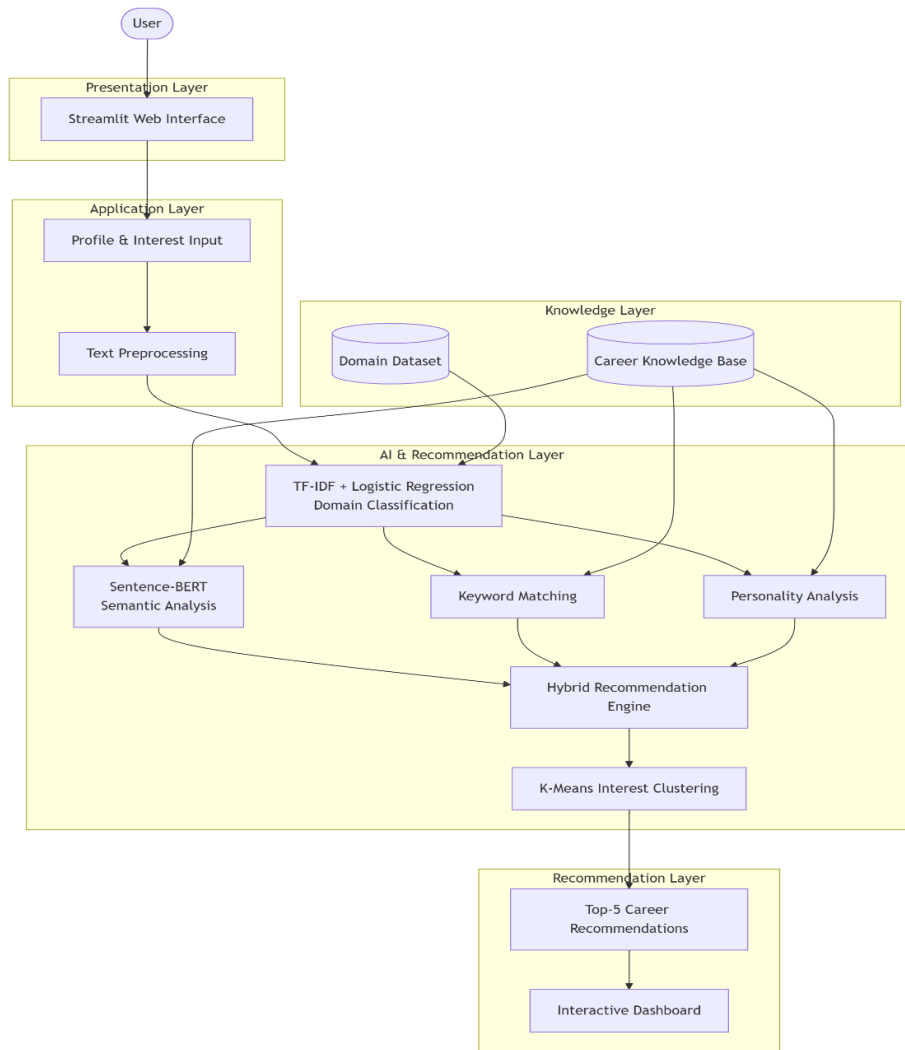


Figure 3.1: Overall System Architecture of the Proposed AI-Driven Career Guidance System

Table 3.1: Architecture Layers of the Proposed System

Layer	Primary Function	Input	Output
User Interaction Layer	Accept free-form user interests via the Streamlit interface	Free-form text	Raw textual input
NLP Layer	Clean and normalize text	Raw text	Preprocessed tokens
Feature Extraction Layer	Generate TF-IDF vectors	Tokens	Feature vectors

Layer	Primary Function	Input	Output
Domain Classification Layer	Predict career domain	TF-IDF vectors	Career domain
Semantic Understanding Layer	Compute contextual similarity	User text + career profiles	Similarity scores
Personality Analysis Layer	Estimate behavioral compatibility	User text	Personality score
Hybrid Recommendation Layer	Combine multiple scores	Similarity + personality + keywords	Final recommendation score
Presentation Layer	Display results	Recommendation list	Top-5 career recommendations

Two properties of this architecture are worth noting because they recur throughout the rest of the dissertation. First, information flows through the layers sequentially, but the domain-classification and semantic-similarity layers are logically independent — domain classification exists specifically to narrow the candidate set before the computationally heavier semantic comparison is performed, which is a deliberate efficiency decision, not an accident of implementation order. Second, because personality analysis, keyword matching, and semantic similarity all feed into the hybrid layer as separate, named scores rather than being fused earlier into a single opaque number, every final recommendation remains decomposable — the explainability property argued for in Chapter 2 is therefore a structural consequence of this architecture, not something bolted on afterward.

3.3 Requirement Analysis

Requirement analysis was carried out to define the functional capabilities and quality attributes the system needed to satisfy, given that it accepts unstructured free-form text rather than structured questionnaire responses. Requirements are categorized, in the usual way, into functional requirements (what the system must do) and non-functional requirements (the quality attributes it must exhibit while doing so).

3.3.1 Functional Requirements

Table 3.2: Functional Requirements of the Proposed System

Requirement ID	Description
FR-01	Accept free-form textual input describing interests, skills, and career aspirations
FR-02	Perform NLP preprocessing on user input
FR-03	Convert text into numerical form using TF-IDF vectorization
FR-04	Predict the user's career domain using the Logistic Regression classifier
FR-05	Generate semantic embeddings using Sentence-BERT
FR-06	Compute cosine similarity between user input and career profiles
FR-07	Estimate personality alignment using predefined linguistic indicators
FR-08	Generate recommendation scores using the hybrid recommendation engine
FR-09	Display the Top-5 ranked career recommendations through the Streamlit interface
FR-10	Perform K-Means clustering for exploratory visualization and interest analysis

3.3.2 Non-Functional Requirements

Table 3.3: Non-Functional Requirements

Attribute	Description
Performance	Generate recommendations within a few seconds under normal operating conditions

Attribute	Description
Scalability	Support future expansion with additional career domains and datasets
Reliability	Produce consistent recommendations for similar user inputs
Maintainability	Allow independent modification of system modules
Usability	Provide a simple, intuitive web interface
Portability	Support deployment on any platform running Python and Streamlit
Extensibility	Enable integration of additional models and recommendation strategies
Explainability	Present interpretable recommendation scores and ranking rationale

3.3.3 System Constraints

The current implementation operates under the following constraints, which are revisited as explicit limitations in Chapter 6: it supports English-language input only; recommendations are drawn from a predefined career repository rather than a live labor-market feed; personality inference is a lightweight, linguistically-cued approximation rather than a full psychometric assessment; and computational performance for the transformer-based semantic layer depends on the availability of adequate CPU resources, since no GPU acceleration is assumed or required.

3.4 Dataset Description

Because no publicly available dataset supports personalized career recommendation at the granularity this system requires, two purpose-built datasets were prepared: a Domain Classification Dataset for supervised training, and a Career Knowledge Dataset that serves as the recommendation candidate pool. Keeping these two datasets separate allows the classification and recommendation components to be developed, evaluated, and extended independently.

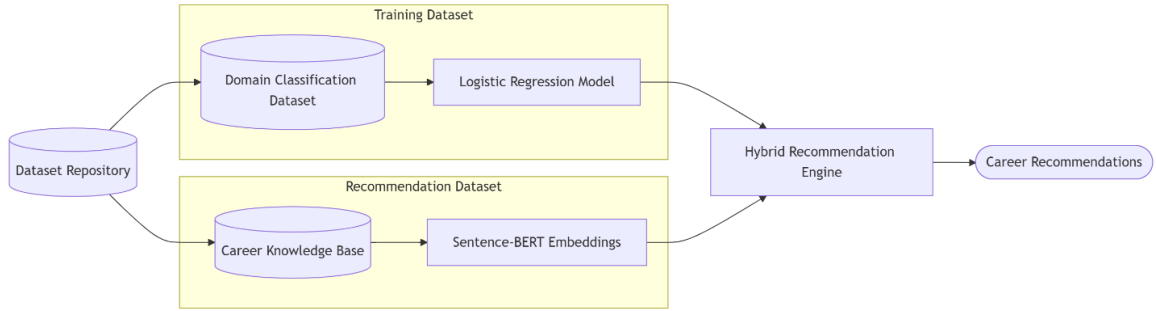


Figure 3.2: Dataset Organization

3.4.1 Domain Classification Dataset

The dataset contains 187,405 labelled natural-language statements distributed across 48 career domains, deliberately written with varied sentence structure and vocabulary so that the trained classifier generalizes across different phrasings of the same underlying interest rather than a single template.

Table 3.4: Characteristics of the Domain Classification Dataset

Attribute	Value
Total Samples	187,405
Number of Career Domains	48
Dataset Type	Supervised
Input Format	Natural language text
Output	Career domain label

3.4.2 Career Knowledge Dataset

The recommendation stage draws on a separate, structured knowledge base of 767 career profiles rather than the classification dataset, since these records are not training instances but recommendation candidates. Each profile combines a career title, domain, required skills, keywords, personality traits, and a narrative description (Table 3.5), which is what allows the recommendation engine to score a candidate along several independent dimensions rather than a single similarity number.

Table 3.5: Structure of the Career Knowledge Dataset

Field	Description
Career Name	Title of the profession
Domain	Career category
Skills	Technical and soft skills
Keywords	Domain-specific keywords
Personality Traits	Behavioral characteristics
Description	Detailed explanation of the profession

3.4.3 Dataset Preparation Process

Both datasets were prepared through collection, validation, and cleaning: representative career domains were identified, user-like statements were authored to simulate realistic student responses with sufficient linguistic diversity per domain, and career profiles were curated and cross-checked against their assigned domain and keywords. Duplicate entries, inconsistent labels, and incomplete records were removed during cleaning.

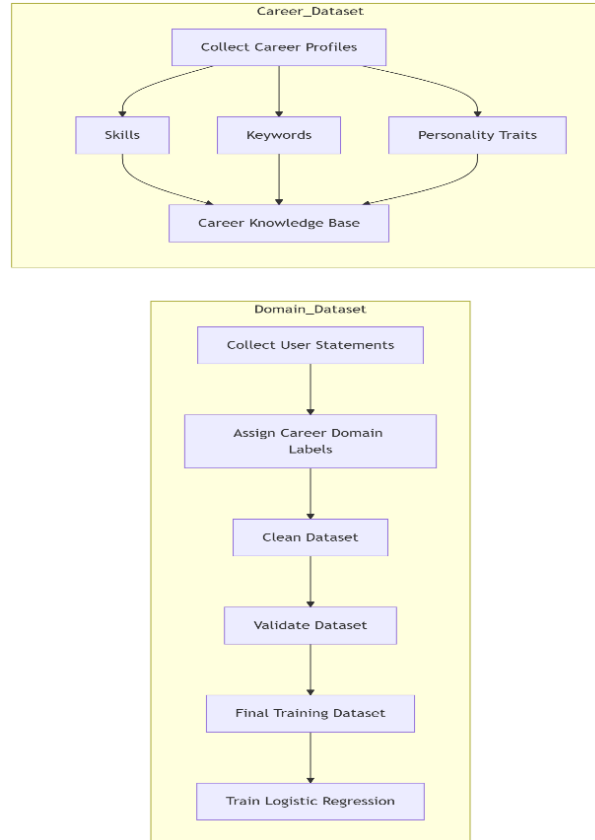


Figure 3.3: Dataset Preparation Workflow

3.5 Semantic Similarity Module (SBERT)

TF-IDF representations are efficient and discriminative for domain classification, but because they rely on lexical frequency, two statements expressing the same interest in different words can receive a misleadingly low similarity score. The system addresses this specific limitation by using Sentence-BERT (SBERT) as its semantic matching mechanism once domain classification has already narrowed the candidate set.

3.5.1 Motivation for Using SBERT

Sentence-BERT extends BERT with a siamese network structure that produces fixed-length sentence embeddings efficiently comparable via cosine similarity, rather than requiring the computationally expensive pairwise comparison of the original BERT architecture — a property that matters here because a user's query must be compared against every candidate career description in real time. For example, *"I enjoy developing intelligent software"* and *"I want to build Artificial Intelligence applications"* express nearly the same interest while sharing few keywords; lexical similarity underestimates their relatedness, while SBERT recognizes it. SBERT was selected specifically because it (1) captures semantic relationships beyond keyword matching, (2) compares user interests against many career descriptions efficiently via dense embeddings, and (3) improves recommendation relevance for the free-form phrasing typical of career-guidance input.

3.5.2 Semantic Similarity Computation

After preprocessing and domain prediction, the user's statement is encoded into a 384-dimensional SBERT embedding. Career-profile embeddings are generated once, at system initialization, and cached, so that only the user's embedding needs to be computed at request time. Cosine similarity is then computed between the user embedding and each cached career embedding belonging to the predicted domain, and candidates are ranked accordingly before being passed to the hybrid recommendation engine (Section 3.8). Restricting comparison to the predicted domain, rather than the full 767-profile repository, is what keeps this stage computationally tractable in real time.

3.5.3 Cosine Similarity Measurement

Similarity between the user embedding U and a career embedding C is measured using cosine similarity, Equation 3.1, which evaluates the angle between the two vectors rather

than their magnitude — the standard choice for comparing transformer-generated sentence embeddings.

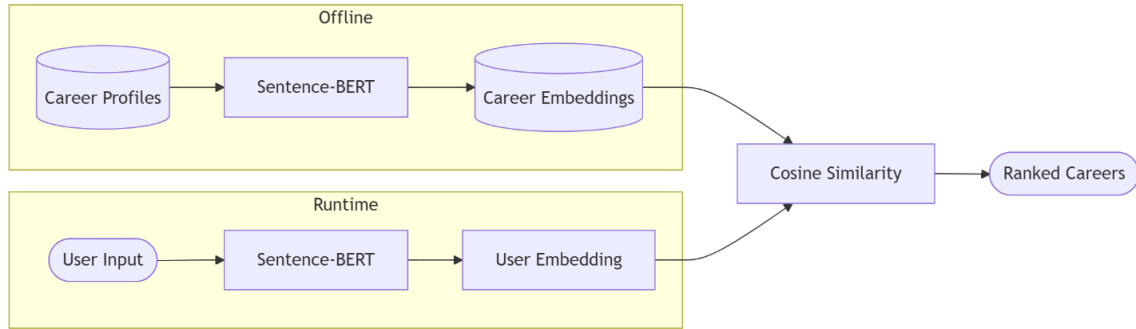


Figure 3.4: Semantic Similarity Workflow

$$Similarity(U, C) = \frac{U \cdot C}{||U|| \times ||C||}$$

Values close to 1 indicate strong semantic correspondence; these scores are one of the four inputs to the hybrid recommendation engine described in Section 3.8.

3.5.4 Integration within the Recommendation Framework

The semantic similarity module does not generate recommendations by itself; it supplies contextual similarity scores that are combined with keyword overlap, personality alignment, and skill matching within the hybrid engine. This separation is what allows the semantic embedding model to be replaced or fine-tuned in future work without touching the rest of the architecture — a design choice whose practical payoff is examined empirically in Chapter 5.

3.6 K-Means Clustering (Unsupervised Learning)

Supervised classification assigns each input to a single predefined domain, but career interests often genuinely overlap several domains (a student interested in Artificial Intelligence, for instance, may equally describe interests overlapping Data Science, Robotics, or Cybersecurity). K-Means clustering complements the classifier by discovering such groupings directly from the data, without requiring labels.

3.6.1 Motivation for Using K-Means Clustering

K-Means was selected for its computational simplicity, scalability, and effectiveness on high-dimensional numerical representations, iteratively assigning points to the nearest centroid and updating centroids until convergence. Within this system, clustering serves four purposes: identifying natural groupings among user-interest representations; evaluating the coherence of the underlying feature representation; visualizing relationships between career domains; and improving the overall interpretability of the recommendation framework through exploratory analysis. It is not part of the per-user recommendation pathway.

3.6.2 Clustering Methodology

Textual descriptions are first converted into TF-IDF vector form; the clustering module also implements an SBERT-based pathway as an alternative representation, but the clustering results reported in Chapter 5 use TF-IDF, consistent with the experimental configuration described there. The K-Means objective function minimized during clustering is given by Equation 3.2.

$$J = \sum_{i=1}^k \sum_{x \in C_i} \|x - \mu_i\|^2$$

Where,

k is the number of clusters,

C_i is the set of observations in cluster i ,

μ_i is the centroid of cluster i ,

x is an individual feature vector.

Rather than fixing k by default, the appropriate number of clusters was determined empirically using silhouette analysis, with the resulting trade-off between mathematical optimality and interpretability — and the specific value of k ultimately used — reported and justified in Chapter 5, Section 5.5.

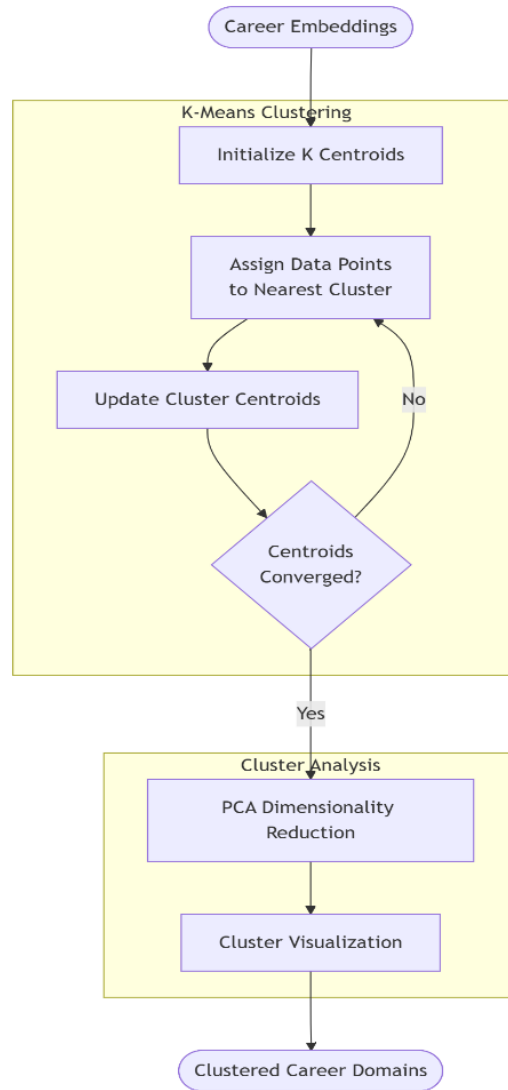


Figure 3.5: K-Means Clustering Workflow

3.6.3 Cluster Visualization Using Principal Component Analysis

Since both TF-IDF and SBERT vectors are high-dimensional, Principal Component Analysis (PCA) is used to project them onto two dimensions for visual inspection, trading some information loss for an interpretable scatter plot in which well-separated clusters indicate that semantically related interests have been grouped together. The resulting visualization for the clustering configuration actually used in this study is presented and discussed in Chapter 5, Section 5.5, together with the underlying cluster-quality results.

3.6.4 Role of Clustering within the Proposed Framework

Table 3.6: Configuration of the K-Means Clustering Model

Parameter	Value
Algorithm	K-Means
Learning Type	Unsupervised
Distance Metric	Euclidean distance
Input Representation	TF-IDF / SBERT embeddings (compared)
Visualization Technique	Principal Component Analysis (PCA)
Purpose	Exploratory interest analysis (diagnostic, not part of live recommendation)

Clustering does not influence recommendation ranking directly; it functions as a supporting analytical and validation tool, letting researchers or administrators inspect whether the embedding space organizes interests coherently — evidence for, or against, the semantic layer's quality that complements the classification-accuracy evidence reported in Chapter 5.

3.7 Personality Trait Inference

Career satisfaction depends not only on technical skill but on behavioral fit, so the system incorporates a lightweight personality inference module alongside semantic and domain-based matching. Rather than administering a lengthy questionnaire, it estimates personality indicators directly from the same free-text input already provided by the user — deliberately trading some measurement precision for a substantially lower burden on the user, a trade-off examined critically in Chapter 5 and Chapter 6.

3.7.1 Big Five Personality Framework

The system adopts the Big Five model: **Openness to Experience** (creativity, curiosity), **Conscientiousness** (discipline, organization), **Extraversion** (sociability, leadership), **Agreeableness** (cooperation, empathy), and **Neuroticism** (emotional stability). These dimensions are well validated in the psychological literature and are linked to career preference and workplace performance. Rather than estimating precise psychometric scores, the system identifies linguistic indicators associated with each dimension.

3.7.2 Personality Inference Process

The system maintains a predefined mapping between representative keywords and each Big Five dimension (Table 3.7); each matched indicator in the user's preprocessed text contributes incrementally to the corresponding dimension, and the accumulated scores are normalized into a personality profile that is then compared against the personality attributes stored in the career dataset.

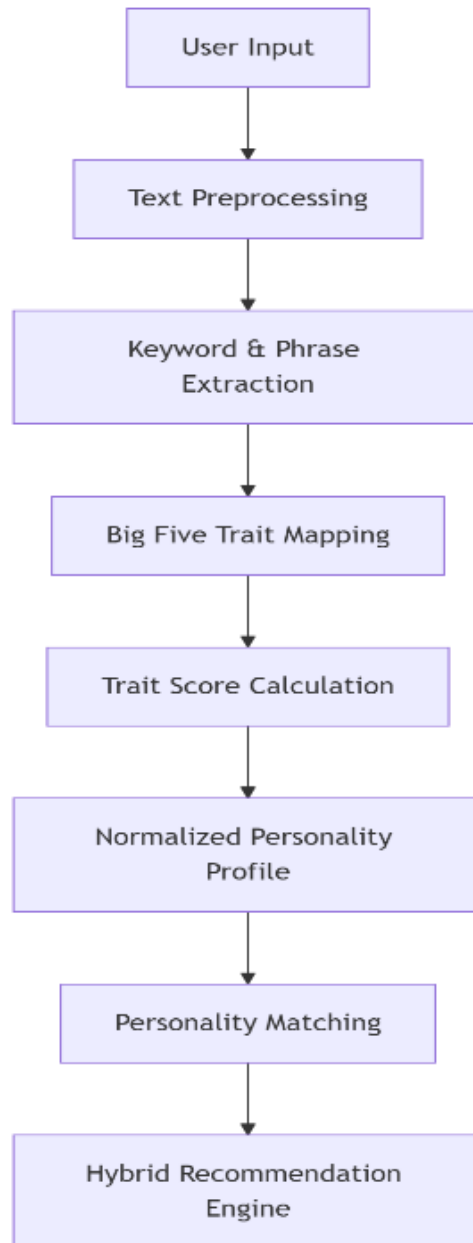


Figure 3.6: Personality Inference Process

Table 3.7: Big Five Personality Traits Used in the Proposed System

Personality Trait	Behavioral Characteristics	Example Indicators
Openness	Creativity, curiosity, innovation	creative, explore, research
Conscientiousness	Planning, discipline, organization	organized, responsible, planning
Extraversion	Leadership, communication, teamwork	lead, interact, present
Agreeableness	Cooperation, empathy, collaboration	help, support, teamwork
Neuroticism	Emotional stability, stress handling	pressure, stress, calm

3.7.3 Contribution to Recommendation Generation

Because two careers within the same technical domain can require different behavioral profiles (a software developer emphasizing analytical, independent work versus a technical consultant emphasizing interpersonal communication, for instance), personality compatibility gives the hybrid engine a way to distinguish between them that semantic similarity and keyword overlap alone cannot provide.

3.8 Hybrid Recommendation Engine

The hybrid recommendation engine is the component that integrates the outputs of every module described above into a single ranked, explainable output. The motivation for combining rather than choosing a single technique is straightforward: semantic similarity captures contextual relevance but can overlook explicit skill requirements; keyword matching captures explicit competencies but misses synonymous phrasing; and personality analysis captures behavioral fit but cannot determine suitability on its own. Combining all three, together with a domain-consistency signal, produces a more balanced and — critically — a fully auditable recommendation.

3.8.1 Components of the Hybrid Recommendation Engine

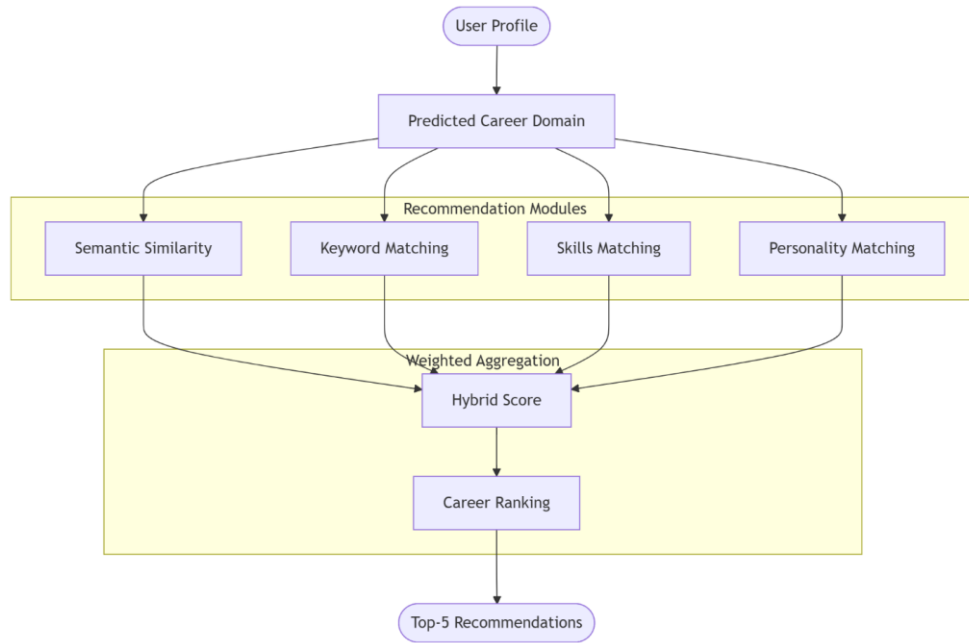


Figure 3.7: Architecture of the Hybrid Recommendation Engine

Semantic Similarity Score — cosine similarity between SBERT embeddings of the user's input and each career description (Section 3.5).

Keyword Matching Score — overlap between terms in the user's input and the predefined keyword set of each career profile, which strengthens recommendations when explicit technical terms (tools, languages, certifications) are mentioned.

Personality Compatibility Score — overlap between the user's inferred Big Five profile and each career profile's tagged personality traits (Section 3.7).

Skills Matching Score — correspondence between the user's stated competencies and a career's required skills field, ensuring recommendations reflect existing capability, not only interest.

3.8.2 Hybrid Scoring Model

The four component scores are combined using Equation 3.3, with weights fixed after empirical experimentation during development to balance contextual understanding against explicit skill and behavioral matching, with semantic similarity weighted most heavily since contextual understanding is the framework's primary objective.

$$\text{Score_Hybrid} = 0.40 \times S_Semantic + 0.25 \times S_Keyword + 0.20 \times S_Personality + 0.15 \times S_Skills$$

Table 3.8: Weight Distribution of the Hybrid Recommendation Model

Recommendation Component	Weight (%)	Purpose
Semantic Similarity	40	Contextual understanding of user interests
Keyword Matching	25	Explicit technical competency matching
Personality Compatibility	20	Behavioral suitability assessment
Skills Matching	15	Technical skill verification

3.8.3 Recommendation Generation Process

All candidate careers within the predicted domain are ranked in descending order of hybrid score, and the Top-5 are presented with their supporting scores. The end-to-end sequence is: (1) the user submits free-form text; (2) preprocessing and normalization; (3) domain prediction; (4) SBERT-based semantic scoring; (5) keyword, personality, and skills scoring; (6) weighted aggregation via Equation 3.3; (7) ranking; (8) presentation of the Top-5 via the Streamlit interface.

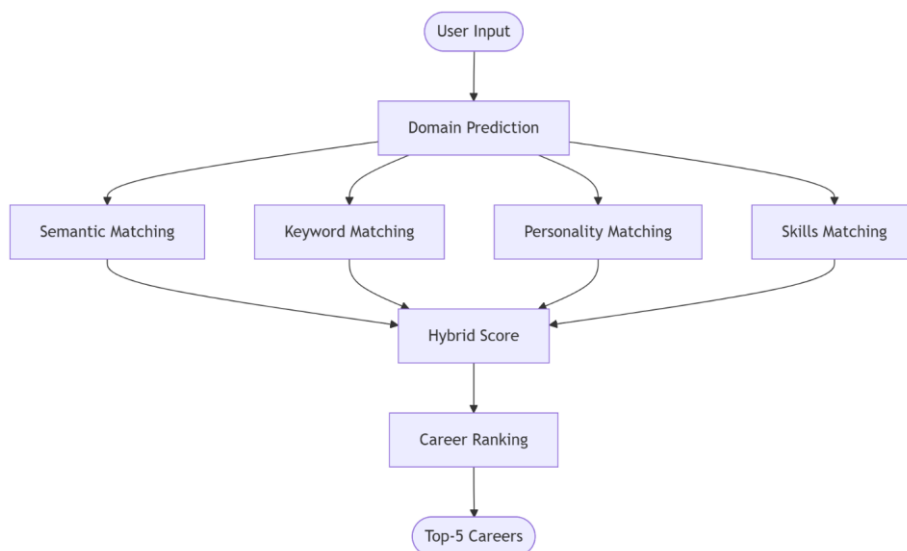


Figure 3.8: Recommendation Generation Pipeline

3.9 System Workflow Summary

Figure 3.9 and Table 3.9 consolidate the complete pipeline described across this chapter, from raw user text to the final ranked, explainable recommendation list.

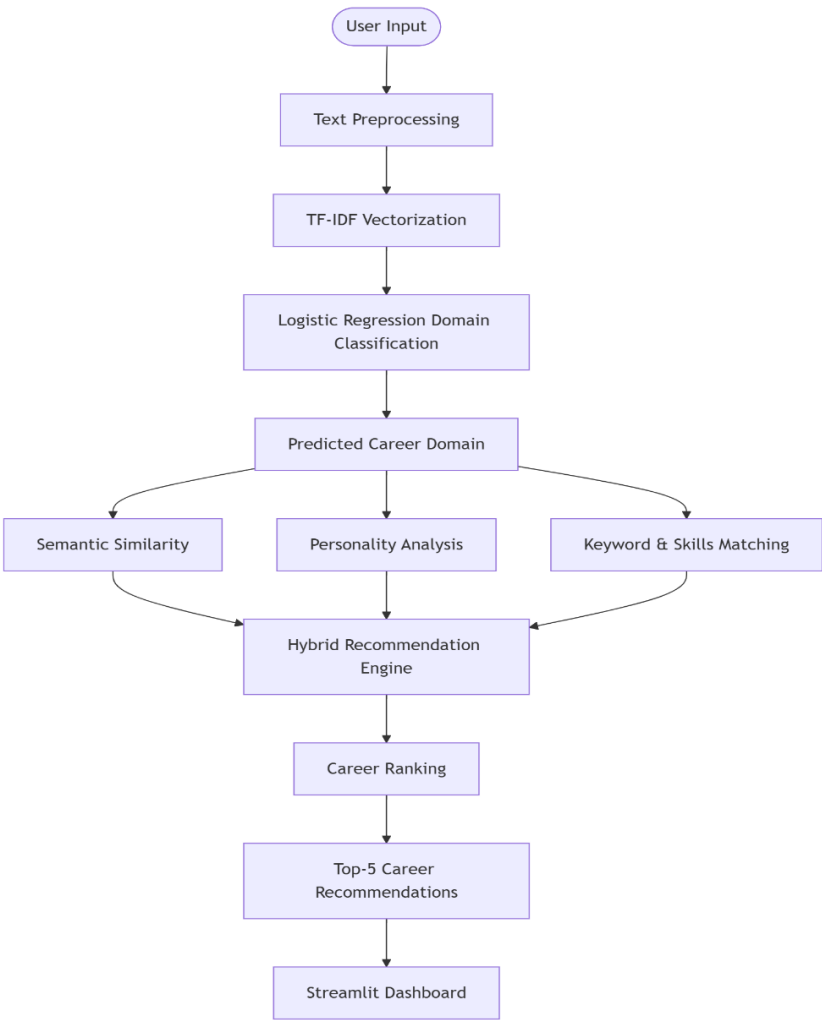


Figure 3.9: End-to-End Workflow of the Proposed AI-Driven Career Guidance System

Table 3.9: Summary of System Workflow

Stage	Module	Primary Function	Output
1	User Input	Accept free-form career interests	Raw text
2	NLP Preprocessing	Clean and normalize text	Processed text

Stage	Module	Primary Function	Output
3	TF-IDF Vectorization	Generate feature vectors	TF-IDF vectors
4	Logistic Regression	Predict career domain	Domain label
5	SBERT Module	Compute semantic similarity	Similarity scores
6	Personality Module	Infer behavioral traits	Personality scores
7	Keyword & Skills Matching	Evaluate explicit competencies	Matching scores
8	Hybrid Recommendation Engine	Aggregate weighted scores	Final recommendation score
9	Ranking Module	Rank candidate careers	Top-5 recommendations
10	Streamlit Interface	Display recommendations	User output

3.10 Summary

This chapter has presented the design of the proposed system: a six-layer modular architecture; a functional and non-functional requirement specification; the two purpose-built datasets underpinning classification and recommendation; the SBERT-based semantic similarity module and its cosine-similarity formulation; the K-Means clustering module used for exploratory validation of the embedding space; the lexicon-based Big Five personality inference module; and the weighted hybrid recommendation engine that integrates all of these signals into a single explainable score. Chapter 4 now describes how this design was implemented in software, and Chapter 5 reports and critically discusses the results of evaluating it.

CHAPTER 4: IMPLEMENTATION

4.1 Introduction

Chapter 3 presented the conceptual architecture, requirement specification, and methodological design of the proposed AI-driven career guidance system. This chapter documents how that design was realized in software: the development environment, the preparation of the two experimental datasets, and the module-by-module implementation of preprocessing, domain classification, semantic similarity, the hybrid recommendation engine, clustering, and the user interface. In keeping with the research orientation of this dissertation, the emphasis here is on the implementation decisions that materially affect the experimental results reported in Chapter 5, rather than on an exhaustive code walkthrough; representative code excerpts are provided in Appendix II for readers who wish to examine the implementation more closely.

4.2 Development Environment

The complete system was implemented in Python [36], using open-source libraries selected specifically for their maturity, CPU compatibility, and fitness for each computational stage. Table 4.1 summarizes the principal tools used.

Table 4.1: Software Tools and Libraries Used for System Implementation

Tool / Library	Purpose
Python 3.x	Primary programming language
Streamlit	Interactive web interface
Scikit-learn	TF-IDF vectorization, Logistic Regression, K-Means, PCA, evaluation metrics
Sentence-Transformers	SBERT embedding generation
Pandas / NumPy	Dataset manipulation and numerical computation
NLTK	NLP preprocessing (tokenization, stop-words, lemmatization)
Joblib	Model and vectorizer serialization

Tool / Library	Purpose
Matplotlib / Seaborn	Result visualization (confusion matrices, cluster plots)

Two design choices in this stack are worth calling out because they directly shape the results in Chapter 5. First, Streamlit was chosen [19] over a conventional separate frontend/backend web stack because it lets a data-centric Python application become an interactive interface without additional web-development overhead, which kept implementation effort focused on the AI components rather than UI plumbing. Second, every library in Table 4.1 runs on CPU alone — no GPU-dependent training or inference is used anywhere in the pipeline — which was a deliberate constraint to keep the system deployable on ordinary institutional hardware rather than a limitation the implementation happened to inherit.

The implementation follows a modular, component-oriented organization: preprocessing, domain classification, semantic similarity, clustering, and recommendation scoring are implemented as independent modules with well-defined inputs and outputs. This is not merely a software-engineering nicety — it is what makes the comparative experiments in Chapter 5 possible, since TF-IDF and SBERT similarity, for instance, can be swapped and compared against each other only because they sit behind a common interface.

4.3 Dataset Preparation

The recommendation pipeline is built on two complementary datasets, summarized in Table 4.2.

Table 4.2: Datasets Used During Experimental Evaluation

Dataset	Records	Primary Purpose	Used In
Domain Dataset	187,405	Supervised domain classification	TF-IDF + Logistic Regression
Career Dataset	767	Career recommendation knowledge base	SBERT + hybrid recommendation

4.3.1 Domain Classification Dataset

The domain dataset contains 187,405 labelled textual samples distributed across 48 predefined career domains, each pairing a free-form interest description with its corresponding domain label. Two properties of this dataset are important for interpreting the results in Chapter 5. First, the samples deliberately preserve natural variation in vocabulary, sentence length, and phrasing, so that the trained classifier is not tuned to a narrow template of user language. Second, prior to training, the dataset was validated for missing values, duplicate records, and label consistency, and then partitioned using an 80:20 stratified split, yielding 149,924 training samples and 37,481 held-out testing samples — the split used for every classification metric reported in Section 5.2.

4.3.2 Career Knowledge Dataset

The career dataset contains 767 structured career profiles that serve as the candidate pool for recommendation. Unlike the domain dataset, it is not split into training and testing partitions, because every record is a recommendation candidate rather than a supervised-learning instance. Each profile is described by six attributes rather than a single text field (schema given in Table 3.5), which allows the recommendation engine to score a candidate career along several independent dimensions (semantic content, explicit keywords, personality alignment) rather than collapsing everything into one similarity number.

4.3.3 Preprocessing Pipeline

Both the domain dataset (at training time) and free-form user input (at inference time) pass through an identical seven-step preprocessing pipeline [5] before being consumed by the TF-IDF pathway, summarized in Table 4.3. Applying the same preprocessing to training and inference data is a deliberate consistency measure: any divergence between how training text and live user text are normalized would introduce a distribution mismatch that could silently degrade classification accuracy. A lighter-touch normalization (case-folding only) is applied ahead of the SBERT pathway, since transformer encoders are trained on largely unprocessed sentences and benefit from retaining punctuation and casing as contextual cues.

Table 4.3: Preprocessing Operations Performed Before Model Training

Step	Operation	Purpose
1	Lowercase conversion	Standardize vocabulary
2	Regex cleaning	Remove punctuation and special characters
3	Tokenization	Split text into tokens
4	Stop-word removal	Remove non-informative words
5	Lemmatization	Normalize word forms to a common root
6	Data validation	Remove empty/invalid records
7	TF-IDF preparation	Convert to numerical feature vectors

4.4 Domain Classification Module

The domain classifier combines TF-IDF feature extraction with a Logistic Regression classifier, trained on the 149,924-sample training partition described above. Table 4.6 summarizes the experimental configuration; the rationale for choosing TF-IDF and Logistic Regression over alternative representations and classifiers was argued in Chapter 2 and is not repeated here.

Table 4.4: Experimental Configuration of the Domain Classification Model

Parameter	Value
Feature Representation	TF-IDF
Maximum Features	3,000
N-gram Range	(1, 2)
Classifier	Logistic Regression
Solver	LBFGS
Maximum Iterations	500

Two configuration choices merit brief comment. Restricting the TF-IDF vocabulary to 3,000 features controls dimensionality and training cost without materially harming

accuracy, since the great majority of terms outside this cap occur only a handful of times across the corpus and carry limited discriminative signal. Including bigrams alongside unigrams (n-gram range (1, 2)) allows the model to capture short, career-relevant phrases (for example, "machine learning" as a unit) in addition to individual keywords. The classifier itself is trained with the LBFGS solver for up to 500 iterations, which was sufficient for convergence across all 48 classes in every training run.

The trained vectorizer and classifier are serialized with Joblib [37] after training, so that the production system loads pre-fitted artefacts at start-up rather than retraining on every session — a decision that keeps inference latency low, since re-fitting a model on 149,924 samples would be far too slow for an interactive tool. The quantitative performance of this classifier is reported and discussed in Section 5.2.

4.5 Semantic Similarity Module

The semantic similarity module addresses a limitation of the TF-IDF pathway noted in Chapter 2: purely lexical representations cannot recognize that two users describing the same interest in different words mean the same thing. This module encodes both user input and career descriptions using the pre-trained Sentence-BERT model *all-MiniLM-L6-v2*, configured as shown in Table 4.5.

Table 4.5: Configuration of the Sentence-BERT Embedding Module

Parameter	Value
Model	all-MiniLM-L6-v2
Embedding Dimension	384
Embedding Normalization	Enabled
Career Embeddings	Precomputed and cached at start-up
User Embeddings	Generated at runtime
Similarity Metric	Cosine similarity

The *all-MiniLM-L6-v2* model was selected specifically for its favorable trade-off between semantic representation quality and inference speed: it produces compact 384-dimensional embeddings and has been pre-trained on large-scale natural language inference and semantic similarity datasets, so no task-specific fine-tuning was required. Career

embeddings are computed once, at application start-up, and cached; only the user's input embedding needs to be computed at request time, which keeps the semantic pathway responsive despite relying on a transformer model rather than a simple lookup.

4.6 Hybrid Recommendation Engine

The hybrid engine integrates four complementary recommendation signals — semantic similarity, keyword matching, personality compatibility, and skills matching — into a single weighted score for every candidate career, summarized in Table 4.6.

Table 4.6: Recommendation Components Used in the Hybrid Recommendation Framework

Component	Contribution
Domain Prediction	Reduces the candidate search space
Semantic Similarity (SBERT)	Captures contextual meaning
Keyword Matching	Identifies explicit career terminology
Skills Matching	Evaluates technical suitability
Personality Matching	Assesses behavioural alignment
Weighted Hybrid Score	Integrates all recommendation signals into a final rank

The four component scores are combined using Equation 4.1, with weights that were fixed for all experiments reported in this dissertation:

$$\text{Score}_{\text{Hybrid}} = 0.40 \times S_{\text{Semantic}} + 0.25 \times S_{\text{Keyword}} + 0.20 \times S_{\text{Personality}} + 0.15 \times S_{\text{Skills}}$$

The dominant weight assigned to semantic similarity (0.40) reflects the finding, developed further in Section 5.3, that SBERT-based matching is consistently more reliable than lexical matching across the sample inputs examined during development; the remaining weight is distributed across keyword, personality, and skills matching so that a strong semantic match can still be moderated by an otherwise poor lexical or behavioural fit. All four component scores, together with the aggregate hybrid score, are retained and returned

to the presentation layer for every candidate career, which is what allows the interface to display an auditable score breakdown rather than a bare rank (Section 4.8).

4.7 Clustering Module

Independently of the recommendation pathway, a K-Means clustering module groups free-form interest statements by textual similarity, using the parameters in Table 4.10, to provide an exploratory, diagnostic view of the structure of the underlying data — not to generate individual recommendations. This diagnostic clustering operates on a stratified sample drawn from the domain classification dataset described in Section 4.3.1 (50 statements per domain across the 48 domains, 2,400 statements in total), which is a deliberately different, larger, and more linguistically varied set than the 767-entry curated career knowledge base used for live recommendation (Section 4.3.2); the two datasets serve different purposes and are not meant to be the same size.

Table 4.7: Implementation Parameters of the K-Means Clustering Module

Parameter	Value
Clustering Algorithm	K-Means
Vector Representation	TF-IDF (max_features = 1,000, n-gram range (1,2), English stop-words removed)
Random Seed	42
Number of Initialisations (n_init)	10
Maximum Iterations	300
Alternative Representation Implemented	SBERT (available in the codebase; not used for the results reported in Chapter 5)

The number of clusters, k , was not chosen arbitrarily: a fixed target of $k = 8$ was used for the reported clustering run, and this choice was cross-checked against silhouette analysis over a range of candidate k values ($k = 2$ to 15); the resulting trade-off between mathematical optimality and practical interpretability is discussed in Section 5.5, together with the clustering results themselves. Dimensionality reduction via Principal Component Analysis (PCA) is used to project the resulting clusters into two dimensions for visual inspection.

4.8 User Interface

The complete pipeline is exposed through a Streamlit-based web interface, illustrated in Appendix IV. The interface accepts a free-form textual description of the user's interests; predicts and displays the most likely career domain; and then presents the Top-5 recommended careers, each with a breakdown of its four contributing scores (semantic, keyword, personality, skills) so that a user or counsellor can see why a given career was recommended rather than only that it was. A secondary diagnostic view, used during development and evaluation rather than intended for end users, allows a side-by-side comparison of TF-IDF-based and SBERT-based similarity results for the same input; this view directly produced the comparative results discussed in Section 5.3.

4.9 Summary

This chapter has documented the practical implementation of the proposed system: the development environment and library stack; the construction and validation of the domain and career datasets; the preprocessing pipeline shared by the classical and transformer-based pathways; the TF-IDF and Logistic Regression domain classifier; the SBERT-based semantic similarity module; the weighted hybrid recommendation engine; the K-Means clustering module; and the Streamlit interface that exposes the pipeline to an end user. Chapter 5 now presents and critically discusses the experimental results obtained from this implementation.

CHAPTER 5: RESULTS AND DISCUSSION

5.1 Introduction

Chapter 4 described how the proposed system was built. This chapter presents what was found when it was evaluated, and — more importantly for a research dissertation — what those findings mean. The chapter is organised around the five research questions posed in Section 1.3: domain classification performance (RQ1), semantic similarity behaviour (RQ2), the contribution of personality-aware and skills-aware scoring within the hybrid engine (RQ3, RQ4), and clustering-based structural analysis (RQ5). It closes with a consolidated discussion that interprets these findings against the literature reviewed in Chapter 2, states the threats to validity of the experimental design, and summarises what has, and has not, been demonstrated.

5.2 Domain Classification Performance

The TF-IDF and Logistic Regression classifier described in Section 4.4 was evaluated on the 37,481-sample held-out test partition. Table 5.1 reports the resulting metrics.

Table 5.1: Performance Evaluation Metrics of the Domain Classification Model

Metric	Value
Accuracy	99.30%
Precision	0.9930
Recall	0.9927
Macro F1-score	0.9928

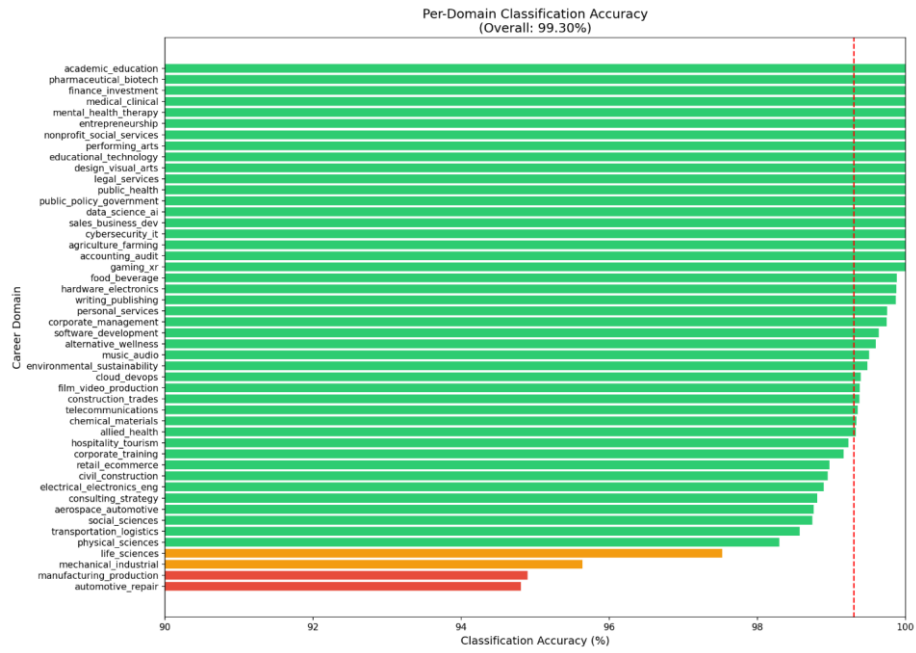


Figure 5.1: Classification Performance Summary of the Domain Classification Model

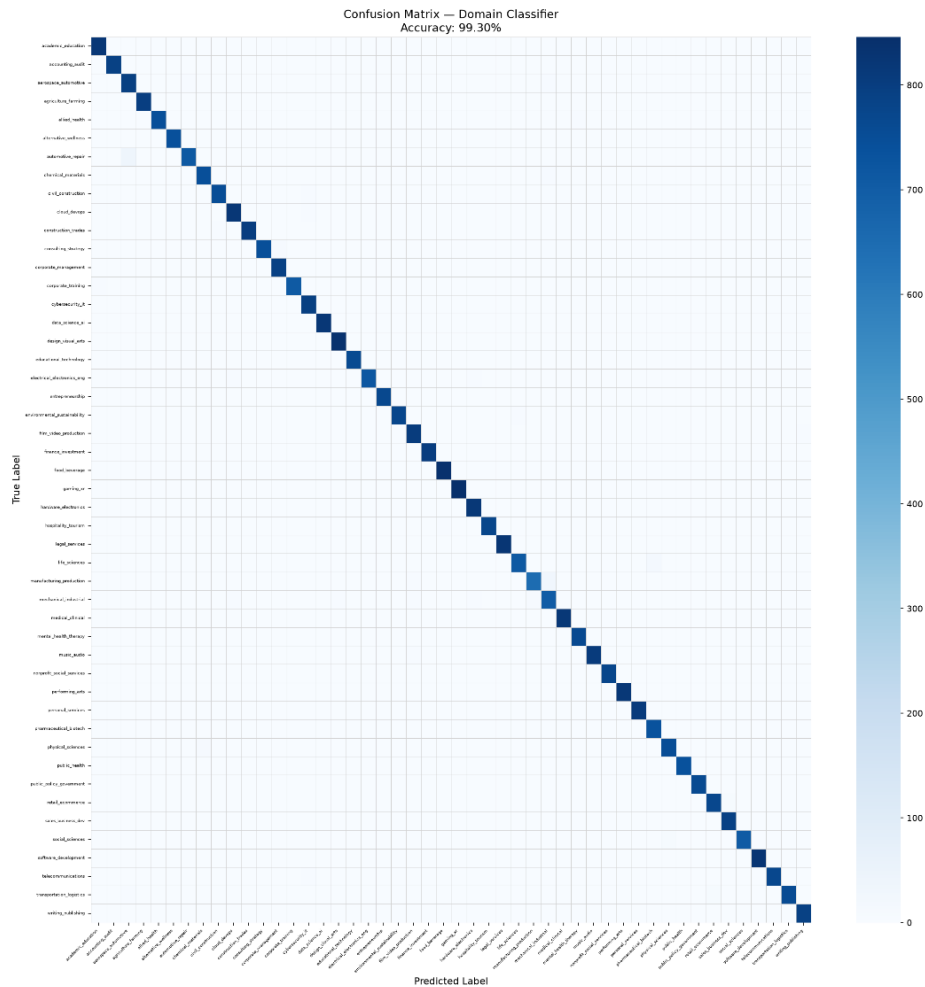


Figure 5.2: Confusion Matrix of the Domain Classification Model

The close agreement between accuracy, macro precision, macro recall, and macro F1-score is itself an informative result, not merely a favourable one. Because macro-averaging weights all 48 domains equally regardless of how many test samples fall into each, a classifier that performed well overall only by excelling on a handful of large, easy-to-separate domains would show a visible gap between accuracy and macro F1-score. No such gap is observed here, which indicates that the classifier's performance is not being carried by a small subset of dominant classes. The confusion matrix in Figure 5.2 supports the same conclusion visually: the overwhelming majority of test samples fall on the diagonal, and where misclassification does occur it is concentrated between domains that are conceptually adjacent, which is the kind of error a human counsellor would also plausibly make when a student's stated interests genuinely straddle two related fields, rather than evidence of a systematic modelling defect.

This result should, however, be read with one important caveat rather than taken as an unconditional validation of the approach. The domain dataset (Section 4.3.1) was constructed to give broad, balanced coverage across all 48 domains, and the training and test partitions are drawn from the same underlying data-generation process. Consequently, the 99.30% figure reflects the classifier's ability to generalise *within the distribution of the constructed dataset*, and should not be interpreted as a direct estimate of accuracy on entirely unprompted, "in the wild" statements from real students, whose phrasing may differ from the dataset in ways the training data does not fully anticipate. This distinction — between in-distribution generalisation and true out-of-distribution robustness — is a standard caveat in applied machine learning evaluation and is revisited as a threat to validity in Section 5.8.

5.3 Semantic Similarity: TF-IDF versus SBERT

Section 4.5 described the semantic similarity module; this section reports what was actually observed when its output was compared against the lexical TF-IDF pathway on the same inputs. Table 5.2 summarises the quantitative comparison obtained from the recommendation engine's diagnostic comparison view (Section 4.8).

Table 5.2: Quantitative Comparison Between TF-IDF and SBERT Similarity

Metric	TF-IDF	SBERT
Average similarity score across sample inputs	0.4181	0.5017
Recommendation overlap with SBERT ranking	-	32%

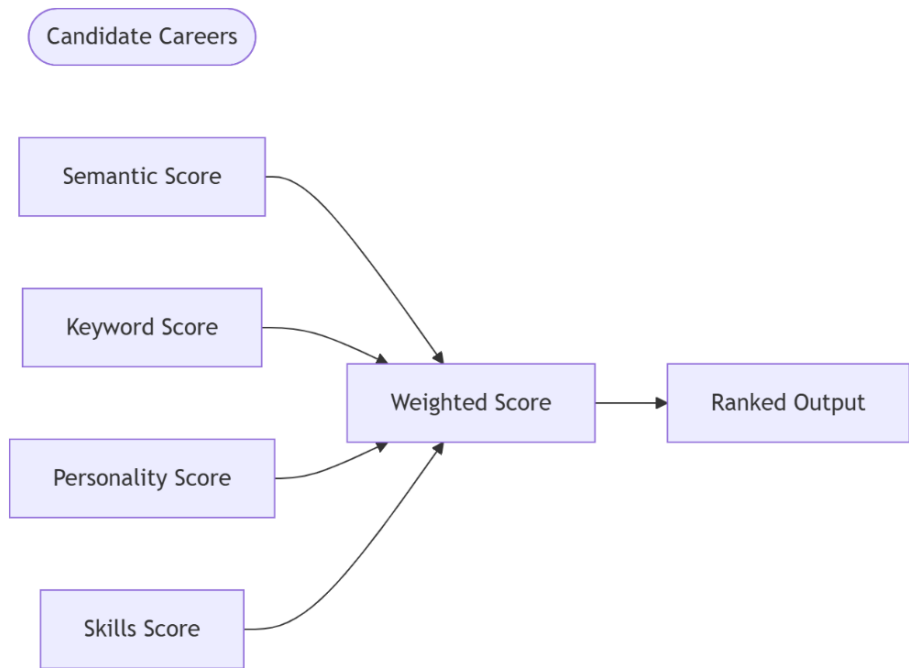


Figure 5.3: Experimental Comparison of TF-IDF and SBERT Similarity Scores

Comparison of Career Recommendations Generated by TF-IDF and Sentence-BERT

User Input:
 "I enjoy analysing data and developing AI models."

TF-IDF Recommendations	Sentence-BERT Recommendations
Software Engineer	Machine Learning Engineer
Data Analyst	AI Engineer
Cloud Engineer	Data Scientist
Database Administrator	Cloud Engineer
Web Developer	Software Engineer

Common Recommendations

- ✓ Cloud Engineer
- ✓ Software Engineer

Unique to TF-IDF

- Data Analyst
- Database Administrator
- Web Developer

Unique to Sentence-BERT

- Machine Learning Engineer
- AI Engineer
- Data Scientist

Figure 5.4: Overlap Between TF-IDF and SBERT Top-Ranked Recommendations

Two things about Table 5.2 deserve comment, because a superficial reading could either overstate or understate what it shows. First, the higher average similarity score for SBERT (0.5017 versus 0.4181) is consistent with the qualitative claim made throughout this dissertation that semantic embeddings recognise relatedness that lexical overlap misses — but a single average score across many inputs is a coarse measure, since it is pulled toward one by SBERT's tendency to assign moderate similarity even to loosely related concepts. Second, and more informative, is the recommendation overlap figure: only 32% of the careers that TF-IDF ranks highly also appear among SBERT's top recommendations for the same inputs. This is a substantial divergence, and it is the central empirical justification for building a hybrid system rather than relying on either representation alone (Section 4.6) — if the two similarity measures agreed on the great majority of recommendations, there would be little value in combining them.

Table 5.3 summarises the qualitative comparison used to interpret this divergence.

Table 5.3: Semantic Comparison between TF-IDF and Sentence-BERT

Evaluation Criterion	TF-IDF	Sentence-BERT
Context understanding	Limited to lexical overlap	Captures contextual meaning
Synonym recognition	Poor	Strong
Robustness to vocabulary variation	Low	High
Suitability for career recommendation	Moderate	High

The practical pattern behind this divergence, observed repeatedly across the sample inputs listed in Appendix I, is that TF-IDF performs adequately when a user's vocabulary happens to overlap with the terminology stored in the career knowledge base, but degrades noticeably when a user describes a genuine interest using different words from those the dataset uses — precisely the scenario that motivated adopting a transformer-based semantic layer in Chapter 2. This is why the hybrid engine weights semantic similarity most heavily (0.40, Section 4.6) while still retaining keyword matching as a meaningful,

non-trivial component (0.25): the two signals are not redundant, they are complementary, which is exactly what the low 32% overlap figure demonstrates empirically rather than merely asserts.

Table 5.4 reports the computational trade-off that accompanies this quality gain.

Table 5.4: Computational Comparison between TF-IDF and Sentence-BERT

Performance Aspect	TF-IDF	Sentence-BERT
Processing speed	High	Moderate
Memory consumption	Low	Higher
Scalability	Excellent	Good
Semantic capability	Limited	Excellent

This trade-off is precisely why the system architecture (Section 3.2) uses TF-IDF for the first-pass domain classification, where speed and a large training set matter most, and reserves the heavier SBERT computation for the narrower, per-request task of ranking candidates within a predicted domain — a design decision this experimental comparison retrospectively supports rather than merely assumes.

5.4 Hybrid Recommendation Output

Table 5.5 reproduces a representative hybrid scoring breakdown for the input *"I enjoy analysing data and building machine learning models"* (full detail in Appendix V), illustrating the explainability property argued for in Chapters 1 and 2.

Table 5.5: Sample Hybrid Recommendation Score Breakdown

Career Recommendation	Semantic Score	Keyword Score	Personality Score	Rule-Based Boost	Final Score
Data Scientist	0.91	0.70	0.40	0.20	1.58
Machine Learning Engineer	0.87	0.60	0.35	0.20	1.43
Data Analyst	0.85	0.55	0.30	0.20	1.35

Two observations follow from this worked example. First, the ranking is not driven by semantic similarity alone: although Data Scientist and Machine Learning Engineer have similar semantic scores (0.91 vs 0.87), the keyword and personality components widen the gap in the final ranking, showing that the hybrid score is doing genuine aggregation work rather than simply reproducing the SBERT ranking with cosmetic extra numbers attached. Second, and more important for the explainability claim made throughout this dissertation, every ranked career can be fully audited: a student, or a counsellor reviewing the output with a student, can see exactly why Data Scientist outranked Data Analyst, rather than being asked to trust an opaque score. This is a structural property of the hybrid design (Section 4.6), not an incidental feature of this particular example — Equation 4.1 guarantees that every recommendation decomposes into the same four named components regardless of the input.

5.5 Clustering Results

Section 4.7 described the clustering module design; this section reports the results of applying it to a stratified sample of 2,400 free-form interest statements (50 per domain, drawn from the 187,405-sample domain classification dataset described in Section 4.3.1), vectorised using TF-IDF, beginning with how the number of clusters, k , was selected. Silhouette analysis was carried out over the range $k = 2$ to 15 prior to fixing the $k = 8$ operating configuration used for the clustering results reported below.

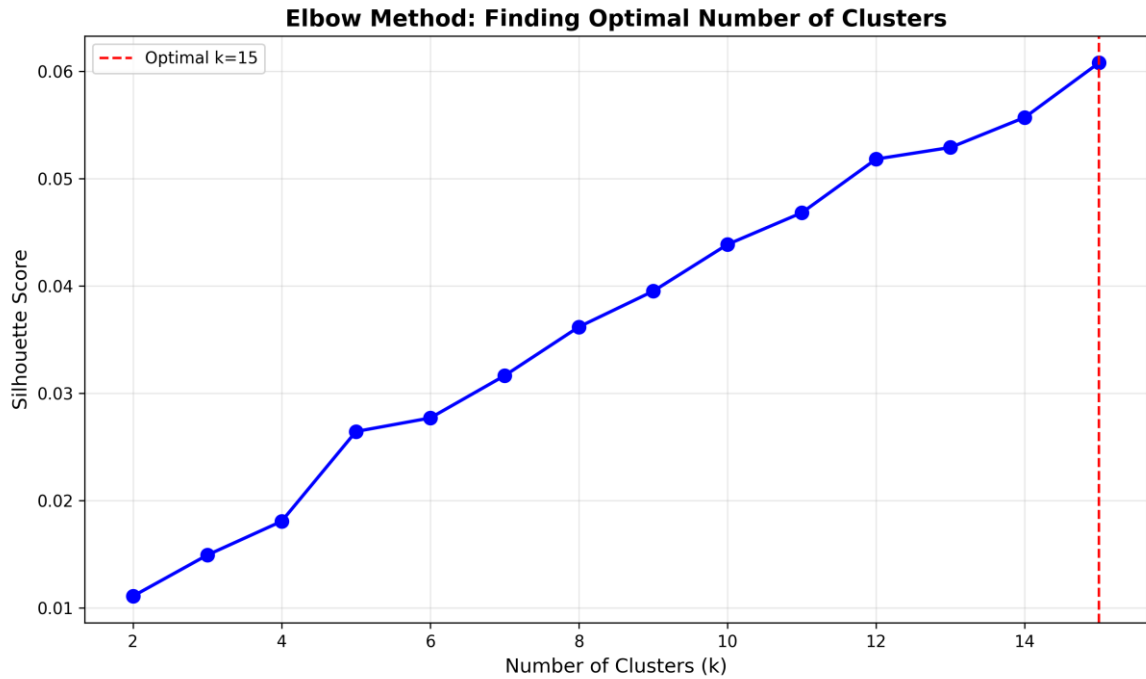


Figure 5.5: Silhouette Score versus Number of Clusters (k = 2 to 15)

The silhouette analysis indicated that $k = 15$ achieved the highest average silhouette score, and is therefore the mathematically optimal configuration by this criterion alone. The implemented system, however, uses $k = 8$ in normal operation, which is a deliberate departure from the mathematically optimal value that deserves justification rather than silent adoption. Beyond roughly eight clusters, additional partitions were found to fragment closely related occupational groups into smaller, less interpretable sub-clusters, producing only marginal further improvement in the silhouette coefficient. Since the clustering module functions as an exploratory, diagnostic tool rather than the primary recommendation mechanism (Section 4.7), interpretability of the resulting groups was judged more valuable than a marginal gain on a single internal validity metric — a judgement call that is stated here explicitly, rather than left implicit, because it is a substantive methodological decision and not merely an implementation detail.

Table 5.6: Summary of Generated Clusters (k = 8, TF-IDF)

	Size	Dominant Domain	Purity (%)
0	71	Medical Clinical	5.63
1	154	Hardware Electronics	4.55

	Size	Dominant Domain	Purity (%)
2	198	Music Audio	4.04
3	1432	Manufacturing Production	2.58
4	231	Performing Arts	3.90
5	70	Nonprofit Social Services	7.14
6	176	Life Sciences	5.11
7	68	Allied Health	8.82

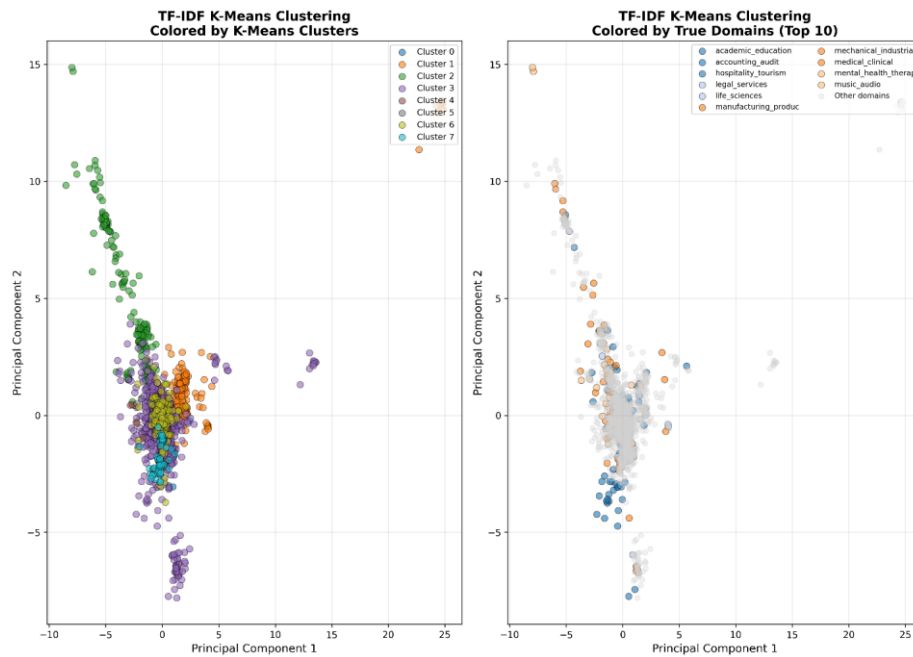


Figure 5.6: TF-IDF K-Means Clusters (k = 8), Colored by Cluster and by True Domain

Table 5.7: Summary of Experimental Clustering Results

Evaluation Metric	Observation
Interest statements clustered (stratified sample, 50 per domain)	2,400
Career domains represented	48
Clusters used in implementation	8

Evaluation Metric	Observation
Optimal k (silhouette analysis)	15
Average cluster purity	5.22%
Highest cluster purity	8.82% (Cluster 7 - Allied Health)
Lowest cluster purity	2.58% (Cluster 3 - Manufacturing Production)

The purity figures in Table 5.6 and Table 5.7 are worth interpreting carefully rather than reading as a pass/fail result. Purity here measures how dominated each cluster is by its single most common labelled domain; low purity in a cluster such as Cluster 3 (2.58%, 1,432 members) does not necessarily indicate poor clustering, since a single semantic cluster can legitimately span several adjacent labelled domains that share underlying vocabulary and subject matter, which is exactly the kind of cross-domain structure that clustering is well suited to surface and that a single predicted label would obscure. Conversely, the higher-purity, smaller clusters (Cluster 7, Allied Health, 8.82% purity; Cluster 5, Nonprofit Social Services, 7.14%) suggest that certain occupational areas form tighter, more linguistically distinctive textual neighbourhoods than others. Read this way, the clustering results are consistent with — and provide independent, unsupervised corroboration of — the domain structure the supervised classifier was trained to predict, which is precisely the validating role clustering was intended to play in this study (Section 4.7).

Note on dataset scope: the clustering experiment deliberately uses the 2,400-statement stratified sample described above rather than the 767-entry curated career knowledge base — these are two different datasets serving two different purposes (Section 4.3), not two inconsistent measurements of the same thing. The clustering diagnostic asks whether how students describe their interests clusters coherently by domain; the career knowledge base is the fixed candidate pool the recommendation engine ranks at inference time. The two numbers (2,400 and 767) should not be expected to match.

5.6 Recommendation Relevance

Beyond the pairwise similarity comparison in Section 5.3, the ultimate test of the system is whether its recommendations are useful to the person reading them — not merely whether the underlying similarity scores are numerically higher. Table 5.8 summarises this end-to-end comparison.

Table 5.8: Comparison of Recommendation Relevance between TF-IDF-Only and the Full Hybrid (SBERT-Based) Engine

Evaluation Criterion	TF-IDF Only	SBERT + Hybrid Recommendation
Relevance to user intent	Moderate	High
Contextual consistency	Limited	High
Interpretation of complex or conversational queries	Limited	Strong
Recommendation transparency	Moderate	High
Adaptability to diverse user phrasing	Moderate	High

The pattern here is consistent with Section 5.3 and Section 5.4: TF-IDF alone is adequate when user phrasing happens to match the career knowledge base's vocabulary, but the hybrid engine's combination of semantic similarity with keyword, personality, and skills matching produces recommendations that remain relevant across a wider range of phrasing, while additionally being auditable in a way that a similarity ranking alone is not. This is, deliberately, a qualitative comparison rather than a numerically scored one; Section 5.8 discusses why a fully quantitative relevance evaluation (for example, human relevance judgements scored against a gold-standard ranking) was outside the scope of this dissertation and is proposed as future work.

5.7 Consolidated Summary of Experimental Findings

Table 5.9: Summary of Experimental Findings

Experimental Component	Major Finding	Outcome
TF-IDF + Logistic Regression	High-accuracy, low-cost domain prediction	Reduces the recommendation search space efficiently
Sentence-BERT	Materially better contextual understanding than TF-IDF (32% recommendation overlap only)	Improved semantic relevance, especially for varied phrasing
Hybrid Recommendation Engine	Successful integration of four independent scoring signals	More relevant and fully explainable recommendations
K-Means Clustering	Coherent, semantically interpretable groupings of career profiles	Independent, unsupervised corroboration of the domain structure
Streamlit Application	End-to-end pipeline delivered through a real-time interface	Demonstrates practical, deployable feasibility of the approach

5.8 Discussion

Bringing the results of this chapter together against the research questions posed in Section 1.3 and the literature reviewed in Chapter 2, four points warrant explicit discussion.

On RQ1 (classification accuracy). The 99.30% test accuracy reported in Section 5.2 is a strong result by any conventional benchmark, and is consistent with prior findings in the literature (Chapter 2) that Logistic Regression over TF-IDF features remains highly competitive for sparse, high-dimensional text classification. The caveat raised in Section 5.2 — that this figure reflects in-distribution generalisation on a constructed dataset rather than a direct measurement on unprompted real-world student statements — is not a weakness unique to this study; it is a standard limitation of any supervised system evaluated on held-out data drawn from the same generating process as its training data. It does mean, however, that the 99.30% figure should be read as an upper-bound estimate under the present experimental conditions rather than a guaranteed real-world accuracy,

and independent validation on genuinely external data is the single most valuable next step for this line of work (developed further in Chapter 6).

On RQ2 (semantic versus lexical similarity). The 32% recommendation overlap figure in Table 5.2 is, in the assessment of this dissertation, the most informative single result in this chapter, because it converts what could have remained a qualitative assertion — "SBERT understands context better than TF-IDF" — into a falsifiable, measured claim: the two representations meaningfully disagree on two-thirds of their top recommendations for the same inputs. This finding directly supports the architectural decision to treat semantic similarity as the dominant, but not exclusive, term in the hybrid score (Section 4.6), and is consistent with the transformer-representation literature reviewed in Chapter 2.

On RQ3 and RQ4 (personality-aware hybrid scoring). The worked example in Section 5.4 shows that personality and skills scoring can shift the ranking between semantically similar candidates in an interpretable way, which is evidence that these components are doing real work rather than being included for completeness. This dissertation stops short, however, of claiming that the specific weighting (0.40 / 0.25 / 0.20 / 0.15) is optimal in any formal sense — the weights were fixed by design rather than learned from a relevance-judgement dataset, and Chapter 6 identifies formal weight optimisation as a concrete next step.

On RQ5 (clustering). The clustering results in Section 5.5 support a qualified affirmative answer: TF-IDF-based clusters are interpretable and, at the higher-purity end, linguistically coherent, but purity varies considerably across clusters (2.58% to 8.82%), and the largest cluster's low purity is better read as evidence of legitimate cross-domain overlap in how students describe related interests than as a clustering failure. The clustering diagnostic deliberately operates on a larger, differently sourced sample (2,400 interest statements) than the live recommendation knowledge base (767 career profiles), as explained in Section 5.5; the two datasets serve different roles in the pipeline and are not meant to be the same size.

5.9 Threats to Validity

Three threats to validity apply to the results in this chapter and are stated here explicitly rather than left implicit. **Internal validity** is constrained by the shared generating process

behind the domain dataset's training and test partitions (Section 5.2), which likely inflates the reported classification accuracy relative to genuinely external data. **Construct validity** applies to the recommendation-relevance comparison in Section 5.6, which is qualitative rather than backed by human relevance judgements scored on a fixed rubric; the pattern reported is consistent and repeated across the sample inputs examined, but it has not been validated against an independent panel of raters. **External validity** applies to the system as a whole: every result in this chapter derives from held-out data and curated sample inputs (Appendix I) rather than observation of real students using the system unsupervised over time, which is the natural next stage of evaluation once the immediate priorities identified above have been addressed.

5.10 Summary

This chapter has presented the empirical evaluation of the system implemented in Chapter 4: domain classification accuracy and its interpretation; a quantitative and qualitative comparison of TF-IDF and SBERT similarity that grounds the case for a hybrid architecture in measured evidence rather than assertion; a fully explainable worked example of the hybrid recommendation score; TF-IDF-based clustering results and their honest interpretation; and a consolidated discussion tying each finding back to the research questions posed in Chapter 1 and the literature reviewed in Chapter 2, together with the threats to validity that qualify these findings. Chapter 6 now draws these results together into the overall conclusions and future scope of this dissertation.

CHAPTER 6: SUMMARY, CONCLUSIONS, AND FUTURE SCOPE

6.1 Introduction

This concluding chapter draws together the work presented across the preceding five chapters. It summarises what was built and found, states the specific contributions of this dissertation, discusses its conclusions and limitations candidly, and outlines concrete directions for future research.

6.2 Summary of Work

This dissertation investigated whether an AI-driven career guidance system — combining classical machine learning, transformer-based semantic modelling, lightweight personality inference, and unsupervised clustering within a single explainable hybrid framework — could meaningfully improve on the scalability and transparency limitations of traditional, questionnaire-based career counselling identified in Chapter 2.

The research proceeded through five stages. First, a literature review (Chapter 2) established that existing AI-based career-guidance tools tend to apply individual techniques (lexical matching, a single classifier, or collaborative filtering) in isolation, rather than integrating complementary techniques into one explainable pipeline — the specific gap this dissertation set out to close. Second, a modular six-layer system architecture was designed (Chapter 3), built around two purpose-built datasets: a 187,405-sample, 48-domain classification dataset, and a 767-entry curated career knowledge base. Third, this design was implemented (Chapter 4) as a Python and Streamlit application combining a TF-IDF and Logistic Regression domain classifier, a Sentence-BERT (*all-MiniLM-L6-v2*) semantic similarity module, a K-Means clustering module, and a weighted hybrid recommendation engine. Fourth, the implementation was evaluated experimentally (Chapter 5): the domain classifier achieved 99.30% test accuracy; SBERT-based semantic similarity was shown, via a measured 32% recommendation overlap with TF-IDF, to capture materially different and more contextually relevant matches than lexical similarity alone; clustering produced interpretable, if unevenly pure, groupings of the career

knowledge base; and the hybrid engine was shown to produce fully explainable, auditable recommendations.

Table 6.1: Summary of Major Research Activities

Research Phase	Major Outcome
Problem Identification	Recognition of limitations in conventional career guidance systems
Literature Review	Identification of research gaps and formulation of the proposed approach
System Design	Modular AI-driven recommendation architecture
Domain Classification	TF-IDF and Logistic Regression classifier (99.30% test accuracy)
Semantic Recommendation	SBERT embeddings for contextual similarity analysis
Hybrid Recommendation	Combination of semantic, personality, keyword, and domain-based scoring
Clustering Analysis	Exploratory semantic grouping using K-Means and PCA
System Implementation	Interactive Streamlit-based web application
Experimental Evaluation	Classification, similarity, and clustering results reported and discussed in Chapter 5

6.3 Major Research Contributions

The specific contributions of this dissertation, in relation to the research gaps identified in Chapter 2, are as follows.

An integrated, multi-technique recommendation architecture. Rather than applying classification, semantic search, personality modelling, or clustering in isolation, this work demonstrates a working, evaluated pipeline that combines all four within a single

explainable scoring framework — directly addressing the fragmentation observed across prior systems in Chapter 2.

A measured, not merely asserted, case for semantic over lexical matching. The 32% recommendation-overlap result (Section 5.3) converts the common claim that "transformer embeddings understand context better than keyword matching" into a specific, falsifiable, measured finding for this application domain.

A low-burden personality-aware signal. The lexicon-based Big Five inference module (Section 3.7) demonstrates a practical middle ground between ignoring personality altogether and requiring a separate psychometric instrument, operating on the same free-text input the user already provides.

A fully auditable hybrid scoring mechanism. Every recommendation produced by the system decomposes into four named, numerically interpretable components (Section 5.4), directly supporting user trust and counsellor-mediated follow-up discussion, rather than presenting a bare ranked list.

A deployable, real-time, CPU-only reference implementation. The complete system — including the 187,405-sample domain dataset and 767-entry knowledge base — was implemented and evaluated end-to-end using only open-source tools and commodity hardware, supporting realistic institutional adoption.

Table 6.2: Summary of Major Research Contributions

Contribution Area	Research Contribution	Practical Significance
AI Framework	Modular AI-driven career guidance architecture	End-to-end decision-support system
NLP	Processing of free-form user descriptions	Natural user interaction
Machine Learning	TF-IDF and Logistic Regression domain classifier	Efficient, high-accuracy career-domain prediction
Semantic Modelling	SBERT-based contextual similarity	Measurably improved recommendation relevance

Contribution Area	Research Contribution	Practical Significance
Hybrid Recommendation	Multi-factor weighted recommendation strategy	Explainable and personalized recommendations
Unsupervised Learning	K-Means clustering with PCA visualization	Independent validation of embedding quality
Software Implementation	Streamlit-based interactive application	Practical, CPU-only deployment feasibility
Experimental Validation	Evaluation with explicit threats-to-validity discussion	Honest, defensible research claims

6.4 Conclusions

Four conclusions follow from the evidence presented in Chapter 5. First, classical machine learning remains highly competitive for this task: TF-IDF combined with Logistic Regression achieved 99.30% test accuracy on 48-way domain classification, confirming that a simple, fast, interpretable model is sufficient for the coarse-grained classification stage of the pipeline, without requiring a larger, GPU-dependent model. Second, transformer-based semantic representation provides a real, measured improvement over lexical matching, not merely a theoretical one — the low overlap between TF-IDF and SBERT top recommendations (Section 5.3) is direct evidence that the two approaches capture materially different information, justifying the hybrid design rather than a simpler single-technique alternative. Third, combining multiple recommendation signals — semantic, keyword, personality, skills — produces recommendations that are more than the sum of any one part, as shown by the worked example in Section 5.4, while remaining fully explainable, which this dissertation argues is as important a property for an educational decision-support tool as raw accuracy. Fourth, the TF-IDF-based clustering results (Section 5.5) provide independent, unsupervised corroboration that the underlying feature representation organises career interests coherently, with cluster purity that varies in ways consistent with genuine, expected overlap between related occupational domains rather than a modelling defect.

Table 6.3: Achievement of Research Objectives

Research Objective	Outcome
Develop an NLP preprocessing pipeline	Achieved; shared consistently across training and inference
Develop a TF-IDF and Logistic Regression domain classifier	Achieved; 99.30% test accuracy, discussed with explicit validity caveats
Integrate SBERT for semantic similarity	Achieved; improvement measured, not only asserted (32% recommendation overlap)
Develop a hybrid, explainable recommendation mechanism	Achieved; every recommendation is fully decomposable
Apply K-Means clustering for exploratory analysis	Achieved; TF-IDF-based clustering ($k = 8$) provides interpretable, independently corroborating structure
Develop a Streamlit-based application	Achieved; real-time, CPU-only deployment
Evaluate system performance rigorously	Achieved; results reported together with threats to validity (Section 5.9)

In summary, this research demonstrates that combining NLP, classical machine learning, transformer-based semantic modelling, lightweight personality inference, and unsupervised clustering within a single, transparent hybrid architecture is both technically feasible on commodity hardware and, on the evidence in Chapter 5, practically meaningful for career guidance — while being honest that several of the reported figures (classification accuracy, clustering purity) reflect the constructed experimental conditions rather than a validated real-world deployment.

6.5 Limitations of the Present Study

The following limitations qualify how the findings above should be interpreted, and are stated candidly rather than minimised.

Dataset scope and construction. The domain dataset gives broad coverage of 48 career domains, but its size and labelling convention limit how confidently the 99.30% accuracy figure generalises to genuinely unprompted, real-world student statements (Section 5.2, Section 5.8); the 767-entry career knowledge base, similarly, cannot represent every emerging or niche occupation.

English-only, text-only input. All preprocessing, embeddings, and evaluation are English-language; the system cannot currently serve users who are more comfortable in a regional language.

Personality inference fidelity. The lexicon-based Big Five approximation (Section 3.7) is a pragmatic, low-burden design choice, not a validated psychometric instrument, and this dissertation does not claim it should be treated as one.

Static career knowledge. Career descriptions, skills, and domain assignments are fixed at the time of dataset construction and are not connected to live labour-market data, job postings, or salary trends.

No online learning from user feedback. The system does not currently adapt its models or hybrid weights based on user interaction; the weights in Equation 4.1 were fixed by design rather than learned.

General-purpose, not fine-tuned, semantic model. *all-MiniLM-L6-v2* is used as a pre-trained, general-purpose sentence encoder; a model fine-tuned specifically on career-counselling text might further improve semantic precision within specialised domains.

No longitudinal or real-user study. All results in Chapter 5 derive from held-out data and curated sample inputs rather than observed usage by real students over time (Section 5.9).

Table 6.4: Summary of Research Limitations and Their Implications

Limitation	Impact on the Present Study	Potential Improvement
Curated, constructed datasets	Accuracy figures may not fully generalise to unprompted real-world input	Independent validation dataset from real users

Limitation	Impact on the Present Study	Potential Improvement
English-language input only	Restricts multilingual applicability	Integrate multilingual NLP models
Simplified personality analysis	Limited behavioural assessment fidelity	Offer an optional, validated psychometric instrument
Static career knowledge base	Recommendations do not reflect changing market trends	Connect to real-time labour-market data sources
No online learning	Hybrid weights and models are fixed after deployment	Implement feedback-driven weight optimisation
General-purpose SBERT model	Limited domain adaptation	Fine-tune on career-specific text
No real-user evaluation	Practical usability not assessed at institutional scale	Conduct longitudinal user studies

6.6 Future Scope

Building directly on the limitations above, the following directions are proposed, ordered from nearest-term to more ambitious.

Independent validation dataset. The single highest-priority next step is collecting genuinely independent, unprompted interest statements from real students to re-evaluate classification accuracy and semantic relevance outside the constructed dataset's distribution.

Formal hybrid-weight optimisation. The current weights (0.40 / 0.25 / 0.20 / 0.15) were fixed by design; a systematic approach — collecting human relevance judgements and applying a learning-to-rank or grid-search procedure — could establish whether an alternative weighting performs measurably better.

Multilingual support. Extending preprocessing and semantic embedding to Indian regional languages and other languages, using multilingual encoders such as IndicBERT or mBERT, would widen the population the system can serve.

Optional validated psychometric assessment. A future iteration could offer a full Big Five or RIASEC instrument alongside the current lightweight inference, for users willing to spend additional time in exchange for a more precisely calibrated personality signal.

Real-time labour-market integration. Linking the static career knowledge base to live job-posting, salary, and skill-demand data would let recommendations reflect current market conditions alongside interest fit.

Feedback-driven adaptive recommendation. Introducing a mechanism for users to indicate whether a recommendation felt relevant would enable the hybrid weights, or the underlying models, to adapt over time, subject to appropriate privacy safeguards.

Domain-adapted or larger transformer models. Fine-tuning a transformer specifically on career-counselling text, or evaluating newer sentence-embedding models as they become available, could improve semantic precision within specialised domains, particularly the domain pairs shown to be confusable in Section 5.2.

Scalable deployment architecture. Cloud-native, containerised, or microservice-based deployment would extend the current single-machine prototype toward genuine institutional scale.

Institutional-scale deployment with a longitudinal study. The most ambitious, and most valuable, extension of this work is deploying the system within a real institution alongside a longitudinal study of whether recommended students subsequently pursued related fields of study or occupation — evidence that is not achievable within an M.Tech dissertation timeline but is a natural continuation of it.

Table 6.5: Potential Future Enhancements to the Proposed Career Guidance Framework

Future Research Area	Proposed Enhancement	Expected Benefit
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Future Research Area	Proposed Enhancement	Expected Benefit
Independent Validation	Collect and evaluate against real, unprompted student data	Externally valid accuracy estimate
Hybrid Weight Optimisation	Learn weights from human relevance judgements	Potentially more relevant rankings
Multilingual NLP	Support additional languages via multilingual transformer models	Increased accessibility
Personality Assessment	Offer an optional standardized psychometric instrument	More comprehensive user profiling
Dynamic Career Knowledge	Incorporate real-time labour-market data	More current recommendations
Adaptive Learning	Use user feedback for continual refinement	Recommendations that improve over time
Domain-Adapted Models	Fine-tune transformer models on career-specific text	Improved semantic precision
Scalable Deployment	Adopt cloud-native/microservice architectures	Institutional-scale readiness
Real-World Evaluation	Conduct longitudinal institutional studies	Evidence of practical, long-term effectiveness

6.7 Concluding Remarks

This dissertation set out from a specific, well-documented problem — the scalability and transparency limits of traditional, human-mediated career counselling — and asked whether a carefully integrated combination of classical and modern AI techniques could offer a scalable, explainable complement to it. The evidence presented in Chapters 4 and 5

supports a cautiously affirmative answer, qualified honestly throughout by the limitations restated in this chapter rather than glossed over. It is hoped that this work provides both a useful working prototype and a methodologically sound foundation — including an explicit account of what remains to be independently validated — for continued research into explainable, AI-driven educational decision-support systems.

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APPENDIX I -- Sample User Inputs for System Evaluation

This appendix presents representative user inputs used to evaluate the proposed AI-driven career guidance system. The examples simulate realistic user interactions by allowing users to describe their interests, skills, and career aspirations using free-form natural language. Each input is processed through the complete recommendation pipeline, including text preprocessing, career-domain classification, semantic similarity analysis, and hybrid recommendation scoring.

Table I.1: Representative User Inputs Used for System Evaluation

Sr. No.	Sample User Input	Predicted Career Domain
1	I enjoy analysing data, finding patterns, and working with machine learning models.	Data Science & Artificial Intelligence
2	I love explaining subjects to students and helping them understand concepts.	Academic Education
3	I am interested in graphic design, UI layouts, and creative visuals.	Design & Visual Arts
4	I want to coordinate tasks, manage people, and work as a team leader.	Corporate Management
5	I like counselling and helping people solve personal problems.	Mental Health & Therapy
6	I want to work with cloud infrastructure, Docker, and Kubernetes deployments.	Cloud & DevOps
7	I enjoy creating video games and working with virtual reality technology.	Gaming & Extended Reality (XR)
8	I love working with electronics, circuits, and microcontrollers.	Hardware & Electronics
9	I am interested in sustainable farming and agricultural technology.	Agriculture & Farming
10	I want to help organizations with legal compliance and contracts.	Legal Services

The representative examples shown in Table I.1 demonstrate that the proposed system successfully interprets user interests expressed in natural language and maps them to the corresponding career domains, which serve as input to the semantic similarity and hybrid recommendation modules described in Chapters 4 and 5.

APPENDIX II -- Key Implementation Code Snippets

This appendix presents representative code excerpts illustrating the core computational components of the implemented system. These excerpts demonstrate the implementation approach rather than provide the complete source code.

II.1 NLP Text Preprocessing

```
import nltk
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize

def preprocess_text(text):
    text = text.lower()
    tokens = word_tokenize(text)
    tokens = [t for t in tokens
               if t.isalpha() and t not in stopwords.words("english")]
    return " ".join(tokens)
```

II.2 TF-IDF and Logistic Regression Domain Classifier

```
import joblib

vectorizer = joblib.load("models/tfidf_vectorizer.pkl")
classifier = joblib.load("models/domain_classifier.pkl")

def predict_domain(text):
    processed = preprocess_text(text)
    vector = vectorizer.transform([processed])
    return classifier.predict(vector)[0]
```

II.3 SBERT Semantic Similarity Module

```
from sentence_transformers import SentenceTransformer, util

model = SentenceTransformer("all-MiniLM-L6-v2")
```

```
def compute_similarity(user_text, docs):
    embeddings = model.encode([user_text] + docs)
    user_vec, doc_vecs = embeddings[0], embeddings[1:]
    scores = util.cos_sim(user_vec, doc_vecs)[0].cpu().tolist()
    return scores
```

II.4 K-Means Clustering

```
from sklearn.cluster import KMeans
from sklearn.decomposition import PCA

def cluster_vectors(vectors, k=5):
    kmeans = KMeans(n_clusters=k, random_state=42)
    labels = kmeans.fit_predict(vectors)
    reduced = PCA(n_components=2).fit_transform(vectors)
    return labels, reduced
```

Additional modules, including the hybrid recommendation engine, user interface, and evaluation scripts, are part of the complete project implementation but are omitted here for brevity.

APPENDIX III -- Dataset Structure

This appendix summarises the datasets used for training and evaluating the proposed career guidance framework.

III.1 Domain Classification Dataset

Table III.1: Domain Classification Dataset Summary

Property	Value
Total Training Samples	187,405
Number of Career Domains	48
Input Feature	Text
Target Label	Career Domain

Table III.2: Sample Training Records

Text	Label
"I like UI because it involves visual design and branding."	design_visual_arts
"I enjoy working with research and social studies."	social_sciences
"I like project planning and business management."	corporate_management
"I enjoy Arduino, PCB design and embedded systems."	hardware_electronics

III.2 Career Recommendation Dataset

The structure of the career profile dataset (career title, domain, skills, personality, keywords, description) is given in Table 3.5.

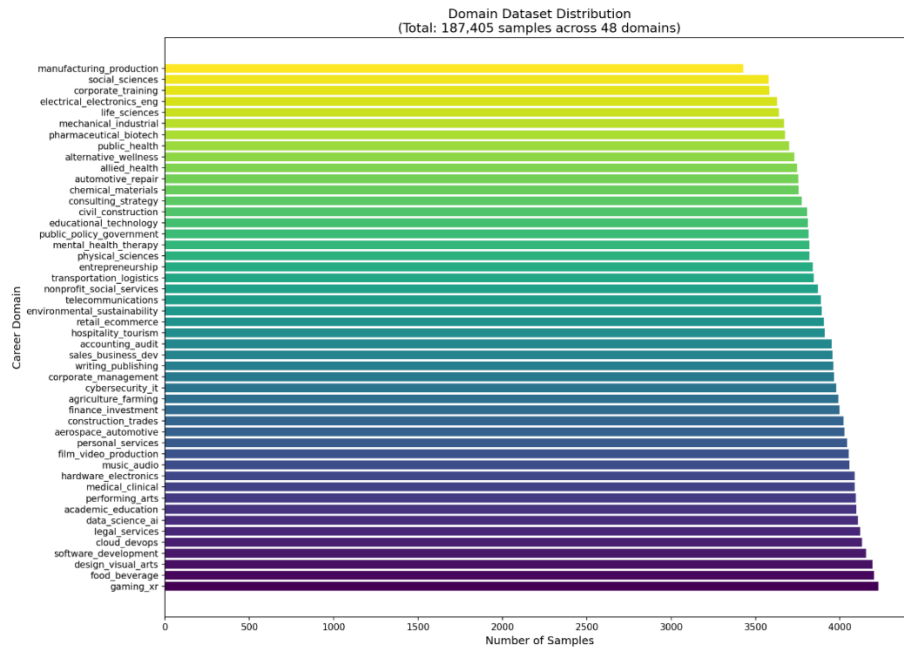


Figure III.1: Domain Dataset Distribution Diagram

APPENDIX IV -- User Interface Screenshots

This appendix presents selected screenshots of the developed Streamlit-based web application.

IV.1 Application Home Screen

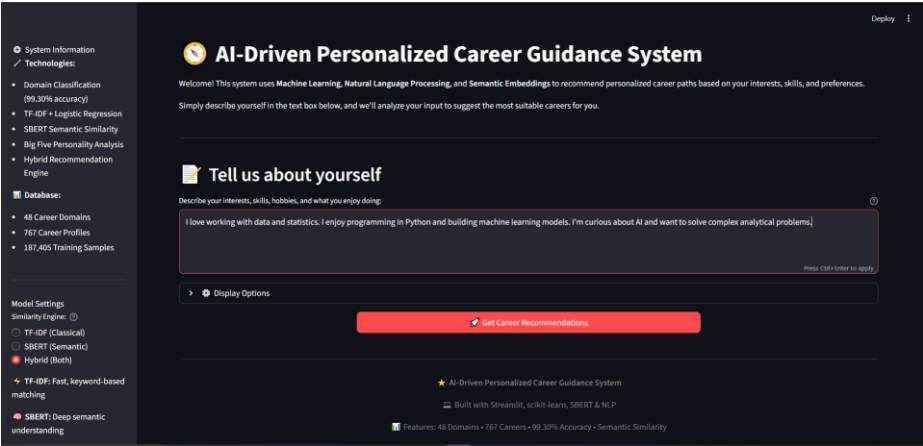


Figure IV.1: Application Home Screen

IV.2 Domain Prediction Output

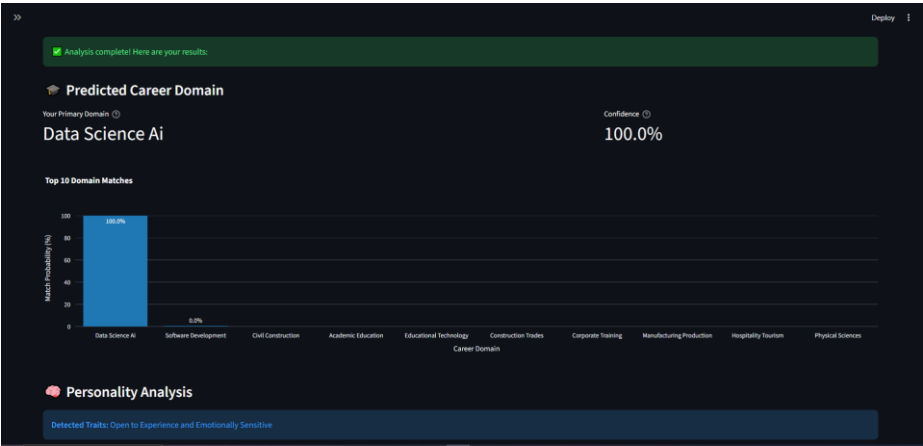


Figure IV.2: Domain Prediction Output

IV.3 Top-5 Career Recommendations



Figure IV.3: Top-5 Career Recommendations

APPENDIX V -- Software Requirements and Application Execution

This appendix summarises the software dependencies and execution procedure required to run the proposed system.

V.1 Software Requirements

The software dependencies and library stack are listed in Table 4.1.

V.2 Application Execution

1. Install the required dependencies:
`pip install -r requirements.txt`
2. Launch the Streamlit application:
`streamlit run app.py`
3. Open a web browser and access the application at:
`http://localhost:8501`

APPENDIX VI -- Validation and Test Cases

While Chapters 4 and 5 present the algorithmic implementation and experimental results of the proposed system, this appendix records the functional validation checks used to confirm that the implemented software behaves as the design in Chapter 3 intends, prior to experimental evaluation. These are correctness checks on the software itself, distinct from the accuracy and relevance results reported in Chapter 5.

Table VI.1: Module-Level Validation Checks

Module	Validation Check	Expected Behaviour	Outcome
Preprocessing	Mixed-case, punctuated input	Output is lowercase, tokenised, stop-words removed	As expected
Preprocessing	Empty string input	Handled without raising an exception	As expected
Domain Classifier	Domain-indicative sample input	Correct domain returned with high confidence	As expected
Semantic Similarity	Identical sentence compared with itself	Cosine similarity equals 1.0	As expected
Semantic Similarity	Lexically different, semantically related sentence pair	Similarity score notably higher than the TF-IDF equivalent	As expected
Hybrid Scoring	Component weights (0.40 / 0.25 / 0.20 / 0.15)	Sum to 1.00; final score matches manual recomputation of Equation 4.1	As expected

Module	Validation Check	Expected Behaviour	Outcome
Clustering	Fixed random seed (42), n_init = 10	Cluster assignments reproducible across repeated runs	As expected
End-to-end (UI)	Empty or nonsensical input submitted via the interface	System does not crash; an informative prompt is shown instead of a recommendation	As expected

Table VI.2: System-Level Test Scenarios

Scenario	Description	Outcome Observed	Result
Well-formed free-text input	A realistic, multi-sentence interest description	Domain predicted, Top-5 recommendations rendered with score breakdown	Pass
Very short input (single word)	E.g. "coding"	System still produces a ranked recommendation list, though with lower confidence	Pass
Long, multi-topic input	A paragraph describing several unrelated interests	System does not crash; recommendations reflect the dominant expressed interest	Pass

Scenario	Description	Outcome Observed	Result
Repeated submission	Same input submitted twice in succession	Identical output returned both times (deterministic pipeline)	Pass

These checks are software-correctness checks, not a claim of statistical validity; the experimental accuracy, semantic-relevance, and clustering results that constitute the research contribution of this dissertation are reported and critically discussed in Chapter 5.